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Kinetically driven synthesis and high-performance Pt-Pd alloy catalysts for PEM fuel cells

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ABSTRACT

The mechanisms of formation of noble metal nanoparticles, particularly their alloys, during liquid-phase synthesis are poorly understood. This is why the synthesis itself remains a combination of science and art, limiting the ability to control the composition and structure of nanoparticles. Researchers are forced to rely on trial-and-error approaches when optimizing methods for obtaining nanostructured materials. We present an original method of liquid-phase synthesis for bimetallic Pt_3Pd nanoparticles based on the sequential introduction of platinum and palladium precursors into a reaction medium under controlled reduction conditions. We demonstrate that the addition of $PdCl_2$ at different steps of the reduction of platinum precursors, corresponding to specific stages of Pt(IV) → Pt(0) transformation, affects the properties of the resulting nanoparticles and Pt_3Pd/C catalysts containing them. Large-scale molecular dynamics simulations allow us to gain insight into the nucleation of bimetallic Pt-Pd nanoparticles under different synthesis conditions. The electrocatalytic behavior of a series of synthesized Pt_3Pd/C catalysts is investigated in oxygen reduction, CO oxidation, and methanol oxidation reactions in an electrochemical cell. The oxygen reduction reaction is also investigated in the membrane-electrode assembly (MEA) of a hydrogen-air fuel cell with a proton-exchange membrane (PEMFC). Some of the obtained catalysts outperform a commercial Pt/C analogue with the same precious metals loading in terms of functional characteristics. According to the results of life tests in MEA (30,000 cycles), the specific power of one of the Pt_3Pd/C catalysts exceeds the specific power of the commercial Pt/C analogue by 28%. The original kinetically controlled liquid-phase synthesis method for bimetallic nanoparticles, validated for the preparation of Pt_3Pd/C catalysts, opens new possibilities for controlling the structure, composition, and properties of a wide range of nanostructured materials. The possibility of using palladium and platinum together for electrocatalyst production could reduce the severity of resource problems caused by limited platinum reserves.

1. Introduction

Noble metal nanoparticles (NPs) and noble metal-based alloys have widespread application in various industries, electronics, and medicine [1–4]. The growing interest in nanostructured materials based on platinum, palladium, and their alloys is largely due to their use as catalysts in various chemical processes, automotive exhaust aftertreatment systems, and electrolyzer and proton-exchange membrane fuel cell (PEMFC) systems [5–10]. Catalysts typically consist of dispersed particles of a support material onto whose developed surface metal NPs are deposited. The composition, structure, size, and spatial distribution characteristics of NPs, as well as their effectiveness in anchoring to the support, determine the activity, selectivity, and durability of catalysts [5,11,12]. Characterization of the resulting nanoparticles is usually

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performed ex situ after synthesis. The need to improve catalyst performance requires the development of new synthesis methods that allow in situ monitoring of the transformation process and better control over their composition and microstructure.

Liquid-phase synthesis methods for noble metal nanoparticles and their alloys are widely used in laboratory and commercial technologies for their preparation. The advantages of liquid-phase synthesis include the relative simplicity of equipment, low energy consumption, and a wide range of possibilities for controlling the process and synthesis outcome, namely, the composition and microstructure of nanoparticles and the supported materials containing them [3,4,13,14]. However, the presence of numerous factors affecting the kinetics and mechanisms of transformations complicates the reproducibility of synthesis, which represents a combination of scientific and empirical components [13, 15–17].

The mechanism of co-reduction of Pt and Pd to form bimetallic Pt-Pd nanoparticles in the liquid phase remains poorly understood. Nevertheless, there is general consensus that the reduction of platinum or palladium precursors to metal nanoparticles is a multistage process involving transformations associated with changes in the composition of metal coordination compounds, step-by-step reduction of metals to the atomic state, formation of small metal clusters, and subsequent nucleation/growth of metal nanoparticles [9,15–19]. It is also noted that the characteristics of the resulting nanoparticles and supported materials are significantly influenced by the adsorption processes of solution components on their surfaces [15,16,18,20–22].

Quinson et al. [15] schematically described the changes in the concentrations of Pt(IV), Pt(II), and Pt(0)x nanoparticles during liquid-phase synthesis: the concentration of Pt(IV) decreases with time; the concentration of Pt(II) passes through a maximum; and the concentration of $Pt(0)_x$ nanoparticles increases non-monotonically from zero at the beginning to a maximum at the end of the reaction. The formation of platinum and palladium nanoparticles may be accompanied by changes in the composition of the initial precursor as a result of interaction with solution components, followed by their subsequent reduction to atoms and the formation of small-sized clusters and nanoparticles [18,21,22]. Recent works [15,17,18,23] emphasize that knowledge of the kinetics of nanoparticle formation and understanding the mechanism of their nucleation/growth are key factors necessary for targeted control over the microstructure of noble metal-based nanoparticles and materials containing these nanoparticles.

Modern research methods make it possible to investigate various stages of precursor reduction and nanoparticle nucleation/growth, but they are typically only applicable to highly diluted solutions. As a result, many conclusions drawn from these studies cannot be directly applied to real technological environments, which are characterized by high concentrations of reagents, medium stirring, and gas bubbling during synthesis [15,16,18,20–22].

The preparation of bimetallic nanoparticles by liquid-phase synthesis methods is a priori more complex compared to that of monometallic ones. This is because the precursors of different metals have different oxidizing powers towards the reducing agents used in synthesis. The kinetics of their reduction may differ, and the interaction of atoms of different elements during nanoparticle formation can facilitate or hinder nucleation/growth processes to varying degrees. Liquid-phase synthesis methods for bimetallic nanoparticles based on noble metals can be conditionally divided into two types: simultaneous reduction and sequential reduction methods [24–26]. Simultaneous reduction of metals from a mixture of precursors often results in the formation of solid solution nanoparticles based on the noble metal components. In contrast, sequential reduction of precursors allows for the preparation of bimetallic nanoparticles either as solid solutions or with an uneven distribution of components [25,27–29]. The synthesis strategy in this case consists of the stepwise formation of nanoparticles with one composition followed by the subsequent decoration of these nanoparticles with atoms of another component(s), i.e., conducting two sequential syntheses in the same reactor.

Herein, we present a novel strategy for controlling the synthesis of binary Pt_3Pd nanoparticles and Pt3Pd/C composites, coupled with a comprehensive analysis of microstructure and catalytic performance towards CO and methanol oxidation, as well as oxygen reduction in acidic media. The principal innovation of this work lies in the in-situ monitoring of metal reduction kinetics, which enables deliberate, temporally-precise introduction of the palladium precursor at predefined stages of platinum reduction. The ability to track reaction progress in situ creates a powerful feedback mechanism, substantially enhancing synthesis controllability and reproducibility in liquid-phase routes to nanostructured materials. This approach is generalizable, offering a versatile paradigm for the synthesis of a broad range of bimetallic and multicomponent materials. In addition, the best of the obtained Pt3Pd/C catalysts demonstrates higher activity and durability compared to Pt/C not only in the electrochemical cell but also in the MEA of the PEMFC.

2. Experimental section

Here is a brief description of the research methods, chemicals and materials. Detailed descriptions accounting for the results and approaches are provided in the Supplementary Material, Section 1.

2.1. Computational details

Classical molecular dynamics (MD) simulations are performed to model the nucleation and growth of platinum and palladium nanoparticles. All simulations are conducted using the LAMMPS software package [30] within a cubic simulation cell of 500 × 500 × 500 Å. The system consists of metal atoms (Pt and/or Pd) and water molecules. Interatomic interactions are described using a combination of potentials: the neuroevolution potential (NEP) [31] for description metal-metal interactions, while the Lennard-Jones potential is used to describe the metal-water (ε = 0.427 eV, σ = 2.466 Å) and oxygen-oxygen (ε = 0.0038 eV, σ = 3.18 Å) interactions [32]. The simulations are carried out in the NVT ensemble at a constant temperature of 350 K, which is maintained by applying a Nosé-Hoover thermostat [33,34] to the metal atoms and water molecules separately. The primary role of the explicit water solvent in this model is to enhance the mobility of metal clusters, preventing them from becoming immobilized and facilitating more frequent collisions, thereby increasing the simulation efficiency. The heat released upon formation of metal-metal bonds is dissipated through a thermostat, which uses a tight coupling constant of 25 timesteps, shorter than the typical 100 timesteps to ensure strict temperature control of the metallic phase in a liquid environment. The equations of motion are integrated with a timestep of 1 fs.

All simulations are performed in a H_2O atmosphere. The initial configurations are as follows: (system 1) pure Pd atomic vapor; (system 2) pure Pt atomic vapor; (system 3) equimolar atomic vapor mixture (Pt:Pd=1:1); (system 4) small Pd seed clusters (4–26 atoms) in Pt atomic vapor; (system 5) medium-sized Pd nanoparticles (70–170 atoms) in Pt atomic vapor.

To calculate the adsorption energies of O_2 on Pt-Pd nanoparticles using density functional theory (DFT) [35], we first optimize the geometry of each studied nanoparticle. The Generalized Gradient Approximation (GGA) is used with the revised PBE [36] parametrization for the exchange-correlation functional as implemented in the VASP software package [37,38]. The ion-electron interaction is described by the projector-augmented-wave (PAW) method [39] with a cut-off energy of 480 eV. The orbital occupancies are smeared using the first-order Methfessel-Paxton method with a smearing width of 0.05 eV. The local geometry optimization of the considered clusters is performed until the maximum force on each atom becomes less than 0.03 eV/Å.

The nanoparticles are placed in a box with a vacuum of at least 10 Å separating them. The spin-polarized Generalized Gradient

Approximation (GGA) with the PBE parametrization is used to calculate the adsorption energies. The Grimme correction (DFT-D3) [40] is applied to account for the dispersive van der Waals interaction. The results of the calculations are post-processed and visualized using the Open Visualization Tool (OVITO) [41] and the VESTA package [42,43]. Electron transfer is studied using the Bader analysis [44].

The adsorption energies of O_2 are calculated using the following formula:

$$E(ads) = E(NP\text{-}O_2) - E(NP) - E(O_2)$$

where $E(NP\text{-}O_2)$ is the energy of adsorption of O_2 on Pt or Pt-Pd nanoparticles, $E(NP)$ is the energy of the considered Pt or Pt-Pd nanoparticles, $E(O_2)$ is the energy of the gas phase O_2 molecules.

2.2. Preparation of Pt_3Pd Nanoparticles and Pt_3Pd/C Materials

Pt_3Pd nanoparticles are synthesized via the polyol method [45]. Ethylene glycol and an aqueous NaOH solution are placed in a chemical reactor. The mixture is heated to 90 °C under an Ar atmosphere. Aqueous H_2PtCl_6 and $PdCl_2$ solutions are then added to the reactor simultaneously with stirring. After the reaction is completed, heating is stopped, and the mixture is allowed to cool to room temperature naturally. To obtain a Pt_3Pd/C material with a 40.2 wt% metal mass fraction, 0.2 g of carbon support are added to the colloidal solution of Pt_3Pd nanoparticles. The product is then filtered, and the precipitates are repeatedly rinsed with bidistilled water and isopropanol. The final Pt_3Pd/C material is dried in an oven at 80 °C and then in a desiccator over P_2O_5 for 24 h. This sample, denoted as p1, corresponds to the co-addition of platinum and palladium precursors at the start of the reaction.

For syntheses involving sequential precursor addition, the procedure is similar to that described above, with the following modifications: the aqueous ethylene glycol mixture initially contains 0.24 M NaOH, and an aqueous 0.035 M H_2PtCl_6 solution is introduced into the reaction mixture. The palladium precursor (an aqueous $PdCl_2$ solution) is then added at a specific time point after the introduction of H_2PtCl_6. To maintain a constant ratio of $S = [NaOH]/([H_2PtCl_6] + [PdCl_2]) = 8$, an additional 5 mL of a 0.15 M aqueous NaOH solution is added concurrently with the $PdCl_2$. The resulting products are designated as p2, p3, and p4, corresponding to different times of Pd precursor addition. A detailed description of the selected addition times can be found in Section 3.2.

A schematic diagram of the chemical reactor used for PtPd nanoparticle synthesis is presented in reference [45].

2.3. Kinetics of transformation of Pt(IV) to $Pt(0)_x$ and Pd(II) to $Pd(0)_x$

2.3.1. RGB-color control

During the reduction of Pt(IV) and Pd(II), which led to the formation of a colloidal solution of NPs in the aqueous ethylene glycol mixture, the color of the reaction medium changed from light yellow to black for H_2PtCl_6 and from maroon to black for $PdCl_2$. This color change is attributable to the reduction of Pt(IV) ions (light yellow) to $Pt(0)_x$ NPs (black), and, similarly, of Pd(II) ions (maroon) to $Pd(0)_x$ NPs (black). A digital microscope was used to record the color of the solution during synthesis. To quantify the coloring intensity of the solution, we used in-house-developed software that converts the color of the solution into three components of the additive RGB color model [46], in which each shade of the continuously recorded image is determined by the contributions of three primary colors: red, green, and blue. This made it possible to graphically interpret the processes occurring in the reaction medium by displaying the change in the quantitative content of each component over time (*Clr vs. t* curves).

2.3.2. Control over changes in the potential of the indicator electrode

During synthesis, we measured the potential of a platinum indicator electrode immersed in the reaction medium (E vs. t curves). For this purpose, the reactor was equipped with a salt bridge, one tube of which was immersed in the reaction mixture and filled with the reaction mixture (before the addition of a chloroplatinic acid solution) [45]. The second tube was filled with a potassium chloride solution and immersed in a beaker containing a solution of the same composition, in which a silver chloride electrode (reference electrode) was placed. The electrodes were connected to a P-40X potentiostat (Elins LTD, Russia), which was used to record the potential difference between the electrodes, corresponding to the redox potential of the medium. The potential values on the chronopotentiometric curves measured during synthesis are reported in this work relative to the standard hydrogen electrode (SHE).

2.4. Characterization Methods for PtPd/C Materials

2.4.1. Determination of Metals Content in PtPd/C Materials

A sample of the synthesized catalyst was placed in a pre-weighed porcelain crucible and calcined at 800 °C for 1 h in a muffle furnace. After cooling, the mass of the crucible containing the incombustible residue (PtPd) was recorded, and the PtPd mass fraction was calculated. The precision of the method was ±1%. A detailed methodology is provided in the Supplementary Material, Section 1.4.1.

2.4.2. X-ray Powder Diffraction Analysis

Measurements were performed on a Bruker D2 PHASER diffractometer using Cu Kα radiation (λ = 1.5406 Å) at a tube voltage of 30 kV and a current of 10 mA. Detailed measurement procedures and the formula used to calculate the mean crystallite size (D_{Av}) are provided in the Supplementary Material, Section 1.4.2.

2.4.3. X-ray Fluorescence (XRF) Analysis

X-ray fluorescence spectra were processed using UniveRS software (Southern Federal University, Rostov-on-Don). The resulting analytical accuracy was ±10%. A detailed methodology is provided in the Supplementary Material, Section 1.4.3.

2.4.4. Transmission Electron Microscopy

The size of the PtPd nanoparticles (D_{NP}) and the characteristics of their size and spatial distributions were investigated by high-resolution transmission electron microscopy (HR-TEM). A detailed methodology is provided in the Supplementary Material, Section 1.4.4.

2.4.5. X-ray Photoelectron Spectroscopy (XPS)

The surface chemical composition of the samples was investigated by X-ray photoelectron spectroscopy (XPS) using a photoelectron spectrometer from SPECS Surface Nano Analysis GmbH (Germany). A detailed methodology is provided in the Supplementary Material, Section 1.4.5.

2.5. Electrochemical Research Methods

Electrochemical investigation methods are detailed in the Supplementary Material, Section 1.5.

2.6. Testing in a Single Membrane-Electrode Assembly

During durability testing of the MEA according to the protocol recommended by the U.S. Department of Energy (DOE), the PGM loading on the cathode (Pt_3Pd/C or JM40) and anode (JM40, Pt/C HiSPEC4000) was 0.4 and 0.2 mg cm^{-2}, respectively. The catalyst layers, with Nafion D520 ionomer (i) at an i/C ratio of 0.7, were formed at 90 °C on a Gore-Select 765.08 membrane by spraying using a manual JAS1186 airbrush. The membrane with the applied catalyst layers was placed between

sheets of Sigracet 22BB GDL on a heating platform, and the assembly was pressed for 3 min at a temperature of 130 ℃ and a pressure of 80 kg cm^{-2}. The catalysts' performance in the MEA was evaluated on a BioLogic FCT-50S test station using a single PaxiTech fuel cell with an active area of 5 cm^2. Air free of mechanical impurities was used as the cathode gas (flow rate: 680 mL min^{-1}), and hydrogen with a purity of 99.999% was used as the anode gas (flow rate: 220 mL min^{-1}). For operation at 100% relative humidity (RH), the gas line temperature and the humidifier temperature were both maintained at about 80 ℃. The gas backpressure was maintained at 1.5 bar throughout.

Procedures for the preparation of catalytic inks and the fabrication and testing of membrane-electrode assemblies are detailed in the Supplementary Material, Section 1.6.

3. Results and Discussion

3.1. Computational Simulations of Synthesis

The formation of nanoparticles from a solution is an extremely complex process in terms of understanding the mechanisms of interaction, diffusion, and segregation of atoms over time. From this point of view, it is important to answer the question of whether the nanoparticles obtained during simultaneous and sequential introduction of platinum and palladium atoms or clusters into the reaction medium differ in composition and structure. Unfortunately, modeling the actual environment of the alkaline polyol synthesis used in the experiments is an intractable task from a computational point of view. Therefore, we performed atomistic simulations of the formation of Pt and Pd nanoparticles in an aqueous environment. We propose five different simplified model systems to investigate different scenarios:

- System 1 consisted of 6000 Pd atoms and 6000 H_2O molecules. The metal atoms and water molecules are initially randomly mixed. The simulation time is 12 ns. This system is designed to study the homogeneous nucleation of pure Pd.
- System 2 consisted of 6000 Pt atoms and 6000 H_2O molecules, initially mixed randomly. The simulation time is set to 12 ns. This setup allows us to investigate homogeneous nucleation in pure Pt and compare its kinetics with that of Pd.
- System 3 consisted of 3000 Pd atoms, 3000 Pt atoms, and 6000 H_2O molecules, all randomly mixed at the start. The simulation time is 12 ns. This model is used to study the co-nucleation of a Pt-Pd alloy from an atomic mixture.

The snapshots during the simulations of System 3 are shown in (Fig. 1a-c). Pd atoms are shown as dark blue, Pt atoms are gray, and water molecules are red.

To model the sequential synthesis strategy used in our experiments, where the Pd precursor is added at different stages of the Pt reduction, we designed two additional systems with staged metal addition:

- System 4 started with small Pd nucleation seeds, with an average size ranging from 4 to 26 atoms. The total number of 6000 Pd atoms is distributed among clusters of various sizes. A minority of atoms are present as single atoms (298), dimers (91 atoms), and trimers (48 atoms), while the overwhelming majority accounting for the remaining 5563 atoms resided in particles consisting of more than three atoms. Platinum atoms are then added to the system in stages: initial 6000 Pt atoms are present at 0 ns, additional 6000 Pt atoms are introduced at 4 ns, and final 6000 Pt atoms are added at 8 ns, resulting in a total simulation time of 12 ns. This setup, containing a mixture of Pd clusters, Pt atoms, and potential Pt-Pd clusters, is designed to simulate a scenario where Pt atoms are gradually released from their salt precursors into a solution containing pre-existing small Pd nuclei.
- System 5 is analogous to System 4, but it features larger pre-existing Pd clusters ranging in size from 70 to 170 atoms. All 6000 Pd atoms are in clusters of more than 3 atoms. The same staged addition of Pt atoms as for System 4 is applied (6000 at 0 ns, 4 ns, and 8 ns). This model aims to investigate the growth mechanisms when Pt atoms are introduced into a solution containing more developed Pd nanoparticles, mimicking the later stage of sequential synthesis.

Analysis of the growth of nanoparticles in Systems 1 and 2 (pure Pt and pure Pd, respectively) revealed that Pd nanoclusters nucleated and grew faster than those of Pt. This is clearly evidenced by a more rapid increase in the average cluster size over time, see red and blue curves in Fig. 1d. This observation is in good agreement with our experimental data. Furthermore, the growth of nanoclusters in the mixed system (System 3) is found to be slower than in the pure Pd system (green curve in Fig. 1d).

Compositional analysis shown in Fig. 1 f determines the atomic percentages of Pt and Pd atoms, both in the bulk and on the surface of nanoparticles. In an equimolar atomic mixture (Pt:Pd=1:1), Pd atoms preferentially segregate to the surface of the particle (Fig. 1 f). This result is consistent with previous theoretical studies [47,48], which indicate that the most energetically favorable structure for bimetallic Pt-Pd nanoparticles is a core-shell configuration with a Pt core and a Pd shell. Therefore, our simulation suggests that the simultaneous synthesis of Pt-Pd bimetallic nanoparticles in an environment containing atoms of both metals will lead to the formation of predominantly core-shell nanostructures with a Pt-rich core and Pd-rich shell.

The remaining Systems 4 and 5 describe the gradual release of Pt atoms into a solution containing small (System 4) and medium-sized (System 5) Pd clusters. Analysis of systems 4 and 5 reveals a nuanced core-shell architecture. While introduced Pt atoms gradually cover the existing Pd cluster seeds. Resulting surface of nanoparticles consistently exhibits a mixed composition (see Fig. 1 g,h). Quantitative analysis showed that, on average, the surfaces of the final nanoparticles contain approximately 34 at% Pd in System 4 and approximately 51 at% Pd in System 5.

Model analysis shows that synthesis method affects atom distribution in bimetallic nanoparticles. If synthesis begins with an environment containing platinum and palladium and proceeds via sequential formation (palladium clusters with platinum atoms), the resulting nanoparticles show different distributions of both atoms in the bulk and on the surface. At the same, the time sequential synthesis approach in which platinum atoms are introduced into a system with pre-formed Pd nuclei does not lead to the formation of a pure Pd surface shell. Instead, this kinetic control strategy allows the formation of unique bimetallic structures characterized by a Pd-rich core and mixed Pt/Pd alloy surface. The presence of Pd atoms on the surface of such nanoparticles can lead to the formation of active sites with an altered electronic structure due to the proximity of Pt atoms. This potentially affects the rate of electrochemical reactions such as oxygen reduction and CO and methanol oxidation.

Ultimately, two aspects are important for the transition to experimental investigation: (1) the process of formation of bimetallic nanoparticles in aqueous-ethylene glycol media at a high pH is distinct from the conditions under which nanoparticles are formed and grown in computational simulations; (2) the laboratory investigation of the electrochemical properties of catalysts containing precious metal nanoparticles and their alloys is carried out following a preliminary electrochemical standardization, which influences the conditions, state, and composition of the surface layers of nanoparticles. Nevertheless, proposed models provide a theoretical basis for the experimental evaluation of the influence of the "platinum subsystem" evolution stage, at which palladium precursor is introduced into the reactor, on the composition/structure of forming nanoparticles. It is also important that the model of sequentially synthesized Pt-Pd/C catalysts has a composite surface, which differs from the ideal core-shell structure often predicted

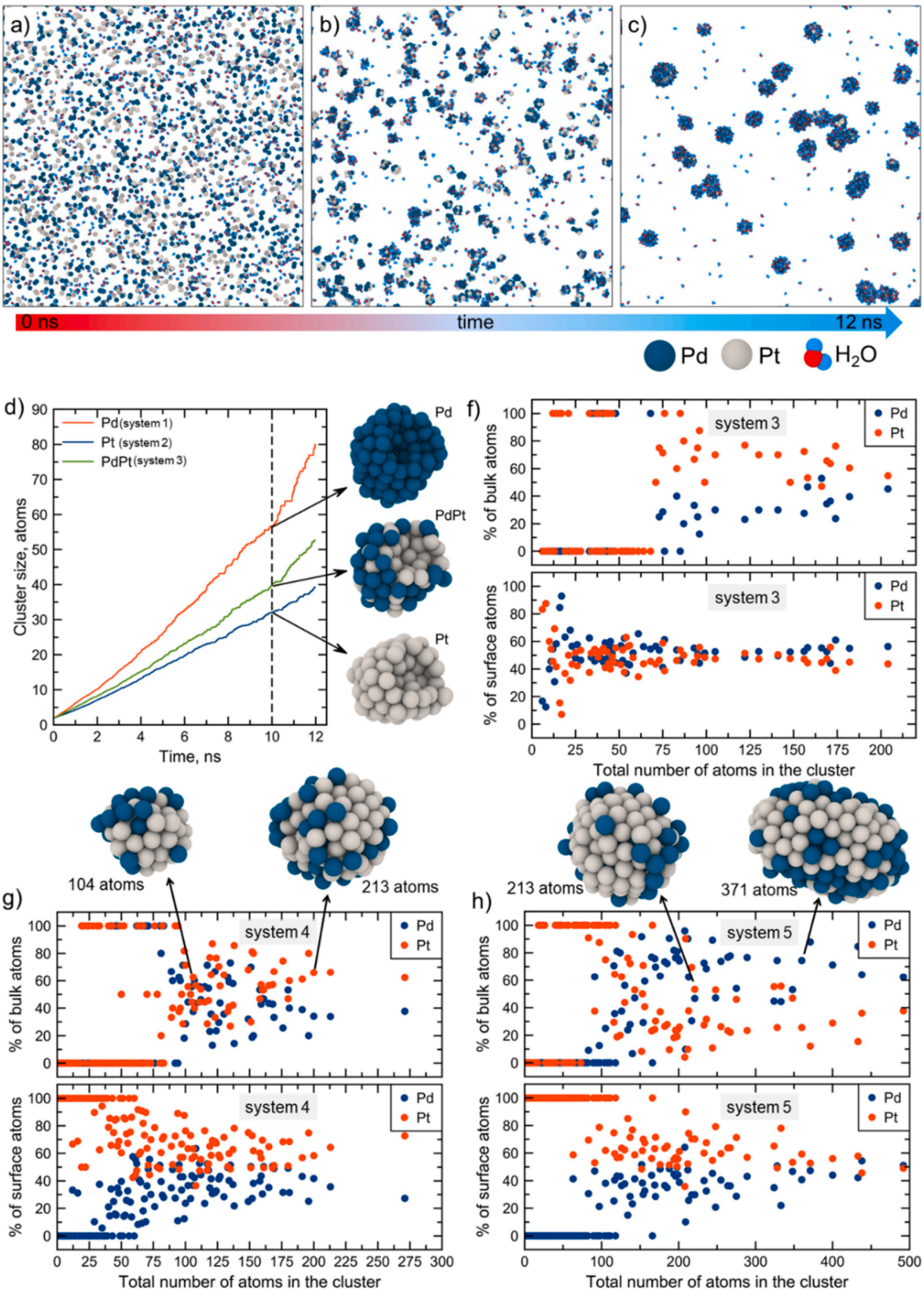

Fig. 1. (a-c) The scheme of the evolution of System 3 containing Pt, Pd atoms, and H_2O molecules over time. Pd atoms are shown as dark blue, Pt atoms are gray, and water molecules are red. (d) Dependence of the average cluster size on time for Systems 1 (red), 2(blue), and 3 (green). Atomic percentage of Pd and Pt in the bulk and on the surface of nanoparticles depending on their size for (f) System 3, (g) System 4, and (h) System 5.

to be a thermodynamically equilibrium structure.

The staged construction of Systems 4 and 5 (pre-existing Pd clusters onto which Pt atoms are introduced in stages) does not reproduce the order in which the precursors are added to the reactor (H_2PtCl_6 first, $PdCl_2$ later); it instead reflects the order in which reduced, zero-valent metal atoms become available to aggregate, which is governed by the markedly different reduction kinetics of the two precursors (Section 3.2; Systems 1 vs. 2, Fig. 1d): Pd(II) is reduced to Pd(0) essentially immediately upon $PdCl_2$ addition, whereas Pt(IV) reduction proceeds through a slower, multistep pathway via Pt(II). Consequently, at the later addition points p2–p4, platinum continues to appear as reduced atoms gradually, over an extended period both before and after $PdCl_2$ is introduced, even though its precursor was added first; this is the scenario approximated by the staged Pt release onto a pre-formed Pd cluster. Systems 4 and 5 are not intended to reproduce the precursor reduction chemistry itself, nor any individual p1–p4 sample quantitatively; their two pre-formed Pd cluster sizes (4–26 and 70–170 atoms) instead bracket the extent of Pd nucleation reached before Pt becomes available, to test whether a composite, mixed Pt/Pd surface remains the qualitative outcome across this range.

To assess whether the mixed-surface architecture predicted for System 5 depends on the specific, computationally motivated solvent density used above, we repeated this simulation at two additional water contents, increasing the number of H_2O molecules by one and two orders of magnitude (60,000 and 600,000, corresponding to $\rho = 0.014$ and 0.14 g/cm^3, versus 6000 molecules, $\rho = 0.0014$ g/cm^3, in the original model), while keeping all other simulation parameters unchanged. A higher water density accelerates cluster growth and promotes coalescence of several particles into agglomerates, yet the fraction of Pd atoms located at the particle surface remains essentially unchanged (45–54%) across this 100-fold variation in solvent density, and no simulation produces a particle fully enclosed by a pure platinum shell; the surface composition remains a Pt/Pd mixture (28–51 at% Pd) at every density tested (Supplementary Material, Section 2, Figures S6–S9, Table S4). This indicates that the mixed Pt/Pd surface architecture predicted by the model is not an artifact of the reduced solvent density adopted for computational tractability.

3.2. Kinetics of Pt_3Pd Nanoparticles Formation

Here, we discuss the kinetics of Pt_3Pd nanoparticle formation during $PdCl_2$ addition at different stages of platinum precursor reduction. Polyol synthesis of Pt_3Pd nanoparticles was conducted at 90 °C in an alkaline solution with an initial concentration ratio S = [NaOH]/

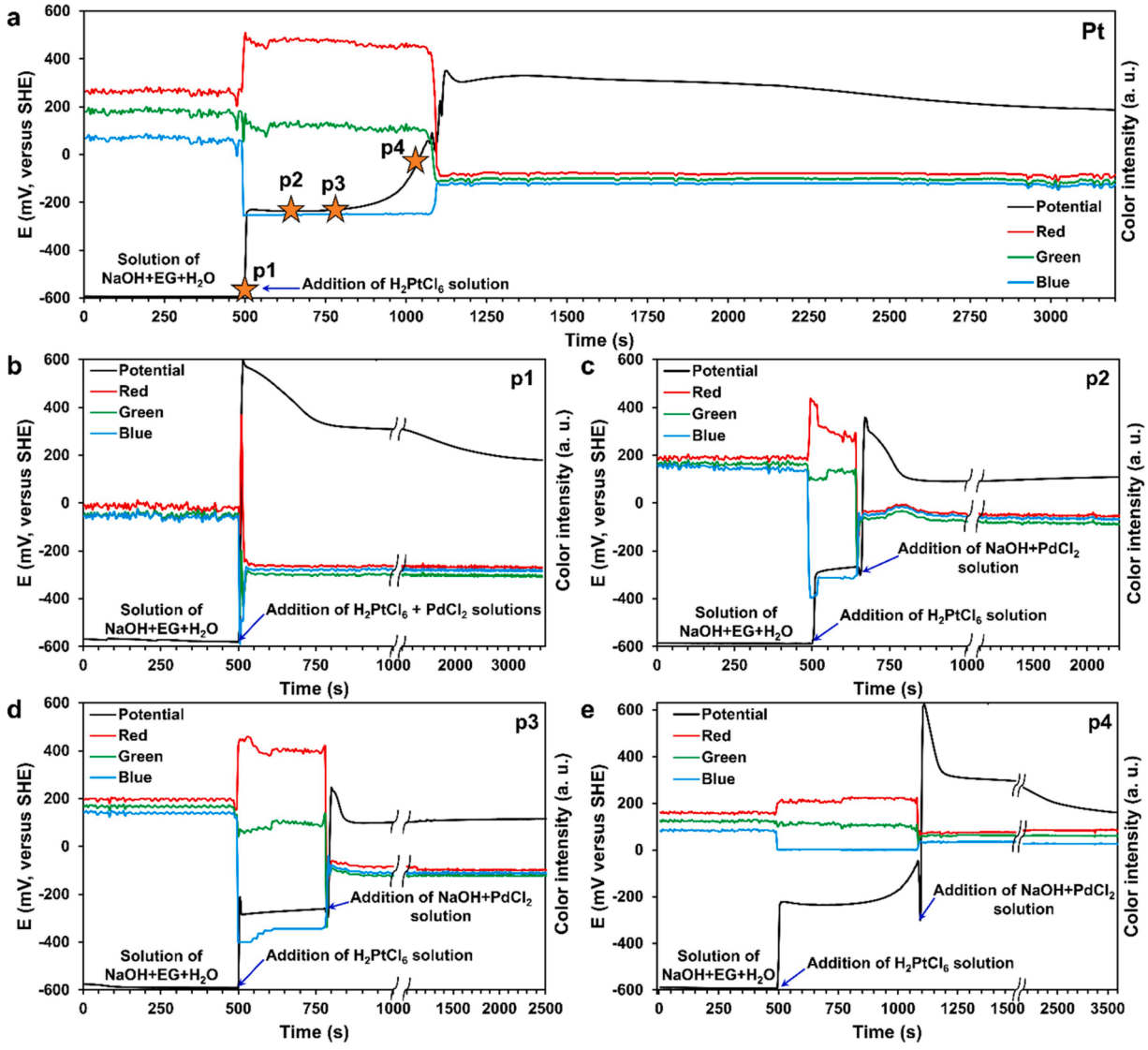

Fig. 2. Temporal evolution of RGB color intensity components (*Clr*): red, green, and blue traces, and redox potential (E): black trace, during H_2PtCl_6 reduction (a) and Pt_3Pd nanoparticle synthesis (b, c, d, e).

$[H_2PtCl_6]$ or S = [NaOH]/($[H_2PtCl_6]$ + $[PdCl_2]$) = 8. Under these conditions, the HCl generated during the synthesis was neutralized to NaCl; however, the reaction medium remained alkaline throughout the synthesis due to the excess sodium hydroxide.

In situ monitoring of the reduction kinetics of platinum and palladium precursors in solutions containing $[PtCl_6]^{2-}$ and mixtures of platinum compounds and $PdCl_2$ was performed by real-time tracking of the solution color and redox potential, as described in [45,49].

Initially, the kinetics of individual H_2PtCl_6 and $PdCl_2$ reduction under analogous conditions were investigated. It was established that the nucleation/growth of Pd nanoparticles commenced almost immediately upon the addition of $PdCl_2$ to the reaction medium. The reduction process of Pt(IV), culminating in the formation of metallic nanoparticles, proceeded much more slowly (Fig. 2a). This difference is partly attributable to the fact that Pt(IV) reduction to Pt(0) proceeds through an intermediate Pt(II) state, whereas Pd(II) can be reduced directly to Pd(0) atoms. This observation corroborates previous molecular dynamics simulations.

Given the differential reduction rates of palladium and platinum precursors, we adopted a strategy of introducing the $PdCl_2$ solution into the reaction medium at different stages of Pt(IV) reduction. The corresponding time points are marked by stars on the *E vs. t* curve (Fig. 2a). Hereafter, the points of $PdCl_2$ addition to the reaction medium are denoted p1, p2, p3, and p4 (Fig. 2a). Addition point p1 corresponds to the simultaneous introduction of H_2PtCl_6 and $PdCl_2$ solutions into a reactor containing an alkaline aqueous ethylene glycol solution (see the Experimental section). Based on the results reported in Ref. [45], the addition points p2, p3, and p4 correspond to states of the system in which the reduction of the platinum precursor has progressed to a lesser (p2) or greater (p4) extent. Additionally, immediately prior to the introduction of $PdCl_2$, an appropriate amount of 1 M NaOH solution was added to maintain a ratio of S = 8.

Fig. 2b–e show the time-dependent evolution of the three RGB color components (Clr vs. t) and the indicator electrode potential (E vs. t), corresponding to the redox potential of the reaction medium, for syntheses carried out with palladium solution addition at points p1–p4. The sharp changes in the intensities of the red, green, and blue color components correspond to: (a) addition of the mixed platinum and palladium precursor solution to the reactor (point p1), and (b) addition of the palladium precursor to the platinum precursor solution at different stages of reduction (points p2, p3, p4), as indicated in Fig. 2.

Based on the characteristic color evolution of the reaction solutions, in particular their darkening, manifested as sharp transitions in all three RGB intensity components on the *Clr vs. t* curves (Fig. 2), the introduction of $PdCl_2$ solution into a reactor containing a platinum precursor solution rapidly initiated the nucleation and growth of metal nanoparticles. For point p1, particle nucleation commenced immediately upon addition of the $PdCl_2$ and H_2PtCl_6 mixture to the reaction medium. Considering previously reported results [45] and the characteristics of the *Clr vs. t* and E vs. t profiles recorded during reaction monitoring (Fig. 2c–e), we can infer distinct chemical states of the platinum compounds at points p2, p3, and p4. Point p2 corresponds to a solution containing a mixture of $[PtCl_6]^{2-}$ and $[PtCl_4]^{2-}$. Point p3 corresponds to a solution containing a mixture of $[PtCl_4]^{2-}$ and Pt(0) atoms [45]. Point p4 is proximal to the onset of platinum nanoparticle nucleation, with platinum existing predominantly as atoms and small atomic clusters [45].

To illustrate the methodology used to select the time point for introducing the palladium precursor into the reaction medium, and to demonstrate the reproducibility of the characteristics of the Pt_3Pd/C catalysts, Table S1 and Figure S1 present the *E vs. t* profiles and selected characteristics of samples obtained from five repeated syntheses of material p4. In these syntheses, the immediate signal for $PdCl_2$ addition was the indicator electrode potential exceeding a value of −100 mV (vs. SHE) (Figure S1). The similarity in the mean crystallite sizes of Pt_3Pd and in the mass activity of the resulting Pt_3Pd/C catalysts in the ORR indicates the reproducibility of their characteristics.

The central question thus emerges: does the chemical state (speciation) of platinum in solution at the moment of $PdCl_2$ addition influence the composition and microstructure of the resulting nanoparticles?

To address this question, nanoparticles are characterized after their deposition onto a carbon support. After synthesis completion, judged by the stabilization of the solution color and redox potential, carbon support material is added to sorb the metal nanoparticles. After 12 h of contact, the metal-carbon composite is recovered, washed, and dried (see the Experimental section). The resulting materials are then analyzed for their composition, microstructure, and electrochemical behavior.

3.3. Composition and Microstructure of Synthesized Pt_3Pd/C Materials

X-ray diffraction patterns of the synthesized materials p1–p4 (Fig. 3) display broadened diffraction maxima characteristic of face-centered cubic (FCC) platinum and palladium. The small difference in lattice parameters between platinum and palladium prevents precise phase attribution; diffraction patterns cannot definitely distinguish between pure platinum, pure palladium, or a palladium-in-platinum solid solution. The mean crystallite size, calculated from the full width at half maximum (FWHM) of the (111) reflection using the Scherrer equation, reveals a size progression: $D_{Av}(p1) < D_{Av}(p2) < D_{Av}(p3) < D_{Av}(p4)$, ranging from 2.2 to 4.0 nm (Table 1).

The metal mass loading in the obtained materials closely matches the theoretical value assuming complete reduction of both metals, yielding 36–41 wt% (Table 1). Considering minor losses during catalyst filtration and washing, essentially complete metal reduction is achieved. Consequently, the resulting metallic phase may comprise either a mixture of monometallic nanoparticles or bimetallic Pt-Pd nanoparticles. X-ray fluorescence analysis of the synthesized materials yields a Pt:Pd atomic ratio of 3.4 ± 0.4 (Table 1), in excellent agreement with the Pt_3Pd/C stoichiometry calculated from precursor quantities.

To clarify microstructural details and elemental composition, including at the individual nanoparticle level, TEM investigations were conducted (Fig. 4, S2-S3, Supplementary Material), combined with elemental mapping and line-scan compositional analysis (Fig. 5).

Examination of the TEM images of the p1, p2, p3, and p4 catalysts reveals a trend of increasing numbers of agglomerates composed of nanoparticles on the support surface, accompanied by a gradual decrease in the specific (per unit surface area) number of particles and an increase in their size from sample p1 to p4 (Fig. 4, S2-S3, Supplementary Material). The mean diameter of individual (non-aggregated) nanoparticles agrees well with the XRD-derived crystallite

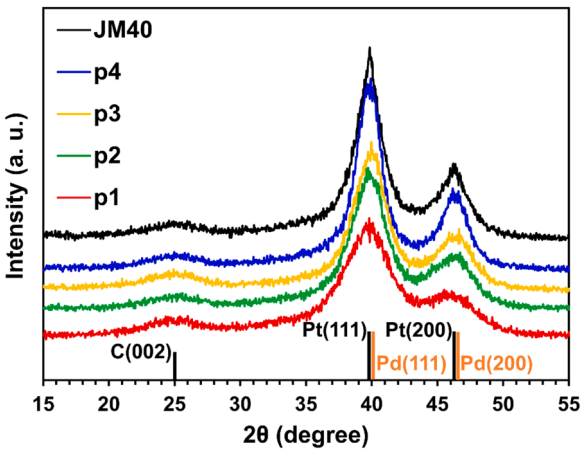

Fig. 3. X-ray diffraction patterns of the synthesized Pt_3Pd/C materials (p1, p2, p3, p4) and the conventional Pt/C catalyst JM40. Cu Kα radiation with λ = 1.5406 Å is used.

Table 1
Composition and Structural-Morphological Characteristics of Catalysts.

Sample	ω(Pt, Pd), % wt.	Pt:Pd (XRF)	Pt:Pd (EDX)	D_{Av}, nm*	D_{NP}, nm**
p1	36.2 ± 0.5	3.0 ± 0.2	3.5 ± 0.4	2.2 ± 0.2	2.7 ± 0.2
p2	39.8 ± 0.5	3.2 ± 0.2	3.4 ± 0.4	2.8 ± 0.2	3.0 ± 0.2
p3	41.3 ± 0.5	3.4 ± 0.2	3.5 ± 0.4	3.1 ± 0.2	3.2 ± 0.2
p4	39.8 ± 0.5	3.0 ± 0.2	3.8 ± 0.4	4.0 ± 0.2	4.4 ± 0.2
JM40	40.0 ± 0.2	-	-	3.5 ± 0.2	3.3 ± 0.2

* D_{Av} – average crystallite size from XRD data
** D_{NP} – average nanoparticle size from TEM data

sizes: D_{NP} increases in the sequence p1 → p4, ranging from 2.7 to 4.4 nm (Fig. 4, S2-S3, Supplementary Material, Table 1). Correspondingly, the size distribution width increases over the same series (size distribution histograms are shown in Fig. 4).

Additionally, HAADF-STEM and SE imaging were performed to clarify the actual location of the NPs on the carbon particles (Figure S3, Supplementary Material). The escape depth of secondary electrons in carbon is about 5 nm [50]. Comparison of HAADF-STEM and SE images from identical regions allows for a qualitative assessment of the proportion of particles on the surface or in the near-surface layer, and also helps to determine whether the observed agglomerates are apparent because of the overlap of NPs from the "top" and "bottom" sides of the support when projecting a three-dimensional object onto a two-dimensional plane. It should be noted that the SE signal, like HAADF-STEM, is sensitive to the atomic number of the element (Z-contrast); therefore, metallic particles down to 1 nm are usually clearly visible in the images against the carbon background. For all samples, a decrease in the number of distinguishable NPs within the same surface area is observed when going from HAADF-STEM to SE images (Figure S3, Supplementary Material). This indicates that a significant fraction of the particles are located either on the "lower" side of the support or within its pores. It is also worth noting that for all materials, fewer agglomerates are observed in SE images compared to TEM and HAADF-STEM images (Fig. 4, Figure S2-S3, Supplementary Material). Nevertheless, the trend of decreasing uniformity of NP distribution over the support surface from p1 to p4 is preserved.

X-ray energy-dispersive spectroscopy (EDX) elemental mapping of surface regions of all synthesized Pt_3Pd/C materials reveals a uniform spatial distribution of Pt and Pd across the carbon support (Fig. 5a, c, e, g). Compositional line-scan analysis confirms that all particles are bimetallic in nature (Fig. 5b, d, f, h), and EDX-based quantification yields a Pt:Pd atomic ratio of approximately 3.5 ± 0.5 for all materials (Table 1).

Minor compositional discrepancies between EDX and X-ray fluorescence results, as well as deviations from the calculated Pt_3Pd/C stoichiometry based on precursor quantities, can be attributed to both the limited accuracy and spatial resolution of EDX analysis and to potentially preferential palladium loss relative to platinum during synthesis, separation, and purification of metal-carbon materials.

Given the substantially higher nucleation rate of palladium compared to platinum, combined with molecular dynamics simulation results, one might hypothesize the formation of nanoparticles with predominantly palladium-enriched cores. However, the compositional line-scan profiles of individual bimetallic nanoparticles (Fig. 5b, d, f, h) do not permit definitive conclusions regarding such a core–shell architecture.

The characteristic features of core-shell nanoparticles include (1) a pronounced concentration gradient at the nanoparticle periphery and (2) relative depletion of the shell-forming component in the nanoparticle interior [25,51]. While we observe the first feature, we do not observe the second. However, the concentration profiles obtained for such small nanoparticles cannot reliably serve as evidence for a homogeneous distribution of both components throughout the nanoparticle volume [52–55].

To address the question of nanoparticle fine structure (architecture), XPS is employed to compare the relative elemental concentrations (atomic ratios) in the near-surface region. Given that there are significant differences in synthesis conditions and mean bimetallic nanoparticle sizes, materials p1 and p4 are selected for XPS investigation, as these metal-carbon materials exhibit the most pronounced differences. Fig. 6 shows the Pt 4 f and Pd 3d spectra of p1 and p4 samples.

Spin–orbit coupling in the Pt 4 f spectra gives rise to two peaks, Pt 4 $f_{7/2}$ and Pt 4 $f_{5/2}$, with a spin–orbit splitting of 3.33 eV and an integrated intensity ratio of 4:3. Literature values for bulk metallic platinum show Pt 4 $f_{7/2}$ binding energies of 71.1–71.6 eV, whereas bulk phases of PtO, PtO_2, and $Pt(OH)_4$ exhibit Pt 4 $f_{7/2}$ binding energies in the ranges 72.3–73.0 eV, 74.0–74.1 eV, and 74.2–74.4 eV, respectively [56,57]. For the investigated catalysts, the Pt 4 f spectrum is adequately described by a doublet (Pt 4 $f_{7/2}$–Pt 4 $f_{5/2}$) with Pt 4 $f_{7/2}$ binding energy of around 71.2–71.3 eV. This doublet is characteristic of platinum in the metallic state.

The Pd 3d spectrum consists of a doublet ($3d_{5/2}$–$3d_{3/2}$) with spin–orbit splitting of 5.26 eV. Literature values for metallic palladium report Pd $3d_{5/2}$ binding energies in the range of 334.8–335.2 eV, whereas palladium in the Pd^{2+} state (PdO) ranges from 336.2 to 336.8 eV, and Pd^{4+} (PdO_2) is centered at 337.9 eV [57–60]. Metallic palladium spectra show no satellite peaks, whereas Pd^{2+} spectra characteristically display satellite peaks at 2.6 and 8.6 eV offset from the main photoelectron peak [61]. All investigated samples exhibit a doublet with Pd $3d_{5/2}$ binding energy of 335.5 ± 0.1 eV, consistent with metallic palladium. This value is slightly higher than that for bulk metallic palladium, a shift that can arise from the interaction of small Pd clusters with the conductive carbon support [60]. For small nanoclusters, charge transfer to the support or the formation of the PdC_x phase, which can arise when carbon is introduced into the near-surface region of Pd, can modulate core-level binding energies [58].

Quantitative XPS analysis of the surface-layer oxygen, platinum and palladium concentrations (normalized to carbon) is presented in Table S2 (Supplementary Material). These results show that the Pt:Pd ratios on the surface of bimetallic samples p1 and p4 are close to each other, but for both samples, significant enrichment in platinum in the near-surface layer is observed compared to its bulk composition. It is reasonable to conclude that palladium atoms segregate preferentially to the nanoparticle core, forming platinum-enriched shells. This configuration is consistent with the observed reduction kinetics.

To summarize, the microstructure and, likely, composition of bimetallic Pt-Pd nanoparticles obtained via polyol liquid-phase synthesis critically depend on the timing of palladium precursor introduction, that is, on the state of reduction and reaction history of the platinum-containing medium. The greater the extent of Pt(IV) reduction at the moment of $PdCl_2$ addition, the larger the resulting mean size of bimetallic nanoparticles. Notably, regardless of the reduction state of the platinum precursor, the introduction of $PdCl_2$ consistently initiates bimetallic nanoparticle formation. The resulting $Pt_{3.4\pm0.4}Pd/C$ materials comprise nanoparticles with a Pd-core–Pt-shell architecture likely resulting from rapid palladium nucleation followed by overgrowth with slower-reducing platinum atoms (or clusters).

3.4. Electrochemical Behavior of Bimetallic Pt_3Pd/C Catalysts in the Oxygen Reduction Reaction (Three-Electrode Electrochemical Cell)

Cyclic voltammograms (CVs) recorded on synthesized bimetallic catalysts and commercial Pt/C are presented in Fig. 7a. CVs differ not only in current magnitude, but also in the shape of the hydrogen adsorption/desorption region spanning the potential range from 0.04 to approximately 0.35 V.

Catalyst p4 exhibits a hydrogen region shape characteristic of Pt/C, with distinct adsorption and desorption maxima for strongly and weakly bound hydrogen, similar to those of JM40 and typical literature data [62]. In contrast, catalysts p1–p3 show atypical hydrogen region profiles

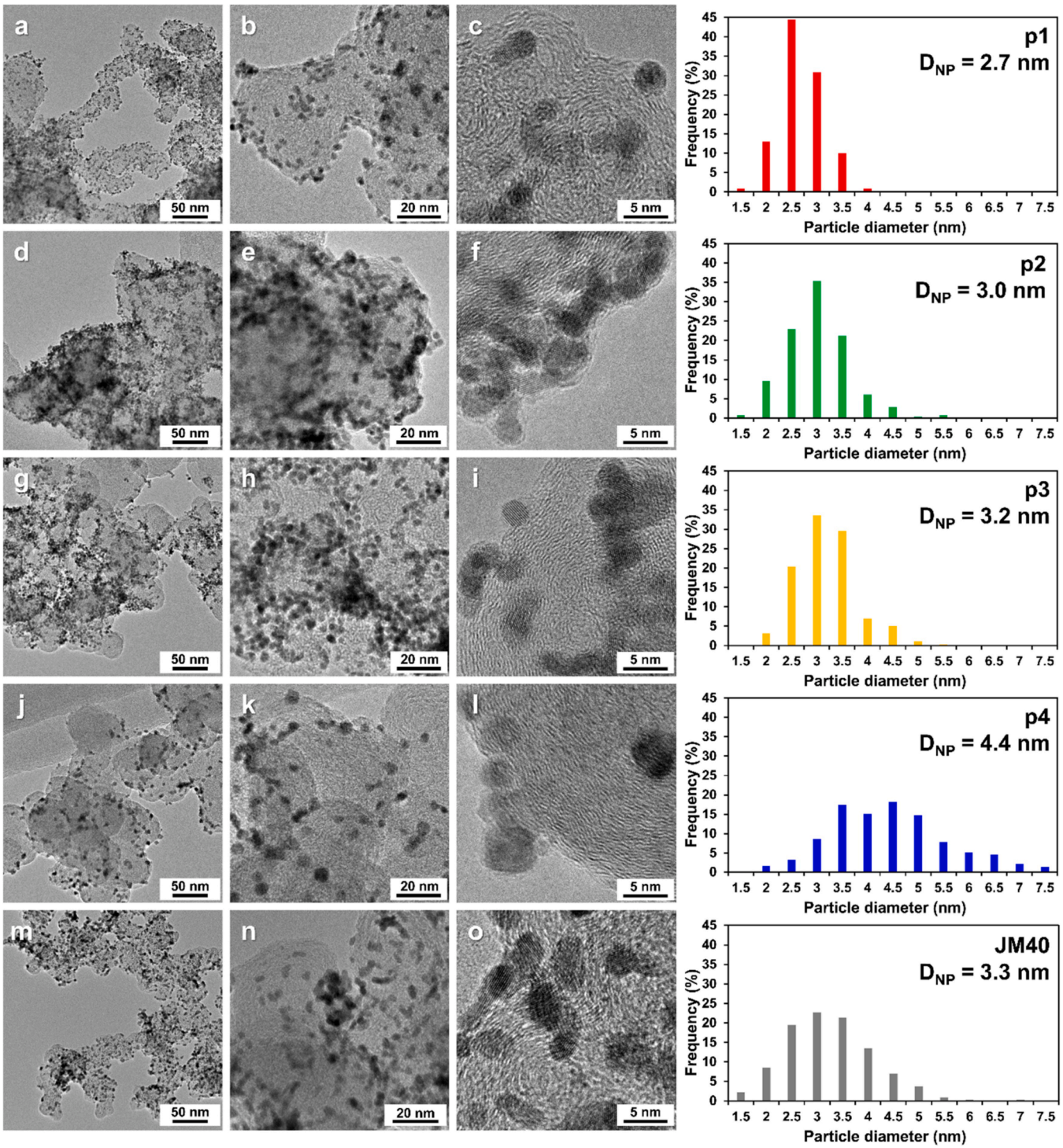

Fig. 4. Transmission electron microscopy (TEM) images of synthesized Pt_3Pd/C catalysts p1, p2, p3, and p4 (a–i) and commercial Pt/C catalyst (m–o), together with corresponding nanoparticle size distribution histograms.

with a single desorption maximum for atomic hydrogen at approximately 0.20 V, presumably corresponding to strongly-bound hydrogen (Fig. 7a). These differences suggest either distinct nanocrystal morphologies in the p1–p3 vs. the p4 series or differences in surface composition and/or defect density. The hydrogen and oxygen region currents decrease across the series p1 > p2 ≥ p4 ≥ p3. Correspondingly, average values of the electrochemically active surface area (ECSA), calculated from the charge associated with atomic hydrogen adsorption/desorption, decrease over this series from 72 to 50 $m^2\ g_{PGM}^{-1}$ (Fig. 7b). This ECSA trend correlates with the increasing nanoparticle size across the same series (Table 1).

It is known that for PtPd/C electrocatalysts, the accuracy of determining ESA based on the charge spent on the adsorption/desorption of atomic hydrogen is questionable due to the possible absorption of part of the hydrogen by palladium [63]. In addition, the magnitude of the charge required to form a monolayer of atomic hydrogen per 1 cm^2 of the true metal surface, in the case of Pt_xPd nanoparticles, may depend on their surface composition, which could vary across the series of p1–p4 materials.

Taking into account the above, we also used a more universal and

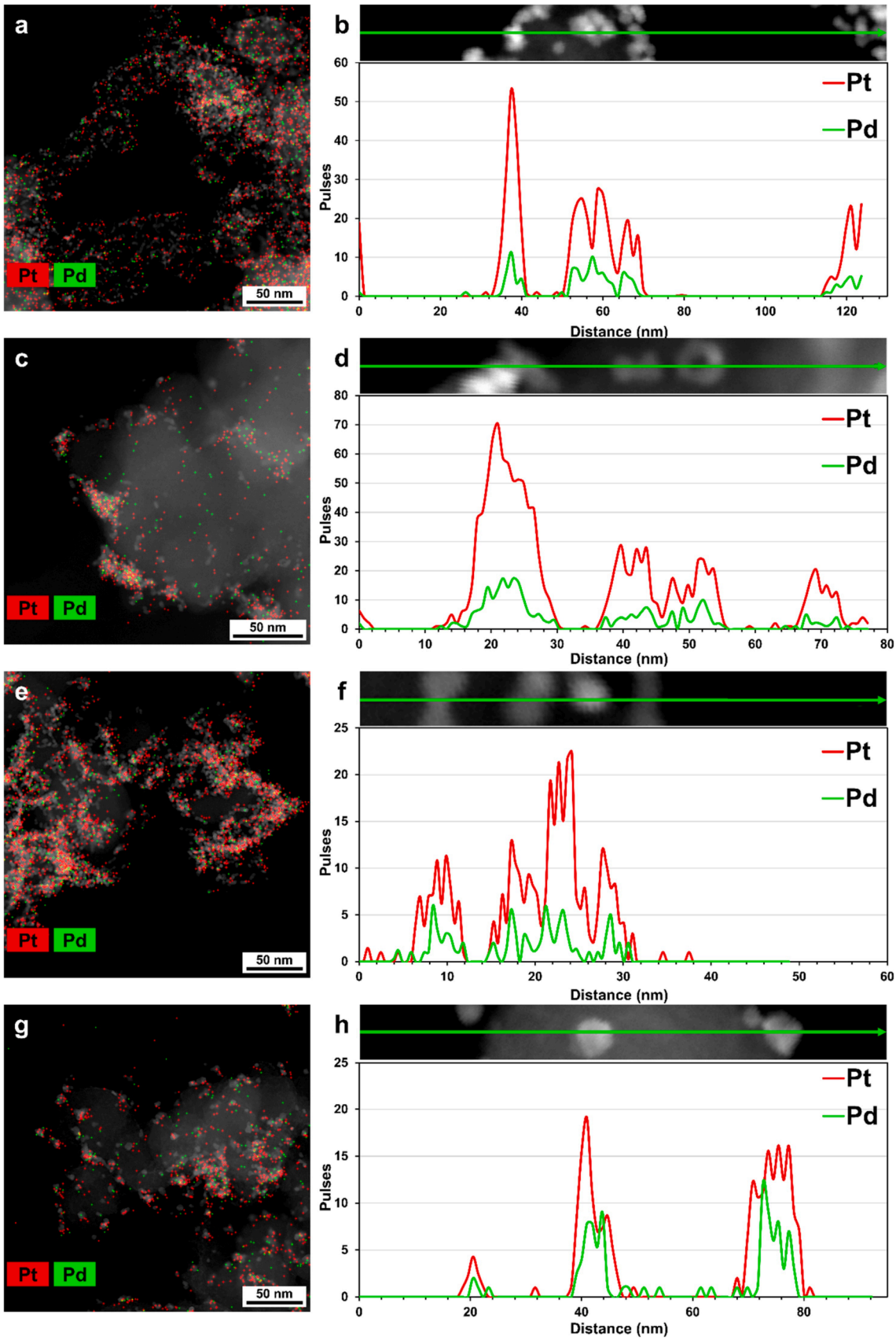

Fig. 5. Elemental mapping of Pt_xPd/C materials corresponding to variants (a) p1, (c) p2, (e) p3 and (g) p4 and corresponding line-scan compositional profiling of the nanoparticles (panels (b), (d), (f), (h) respectively).

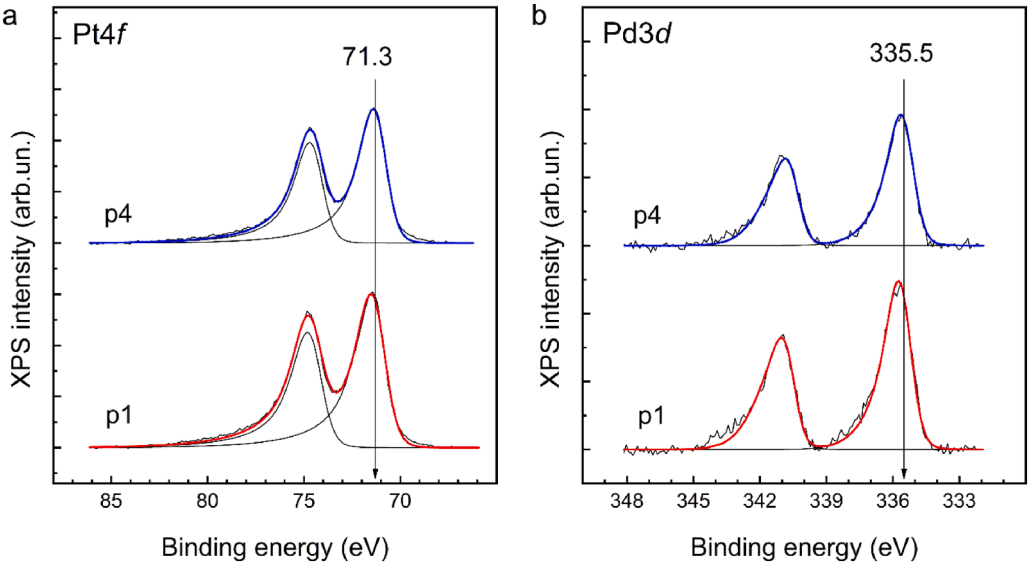

Fig. 6. XPS spectra of (a) Pt 4 f and (b) Pd 3d for p1 and p4 variants of Pt_xPd/C material. Spectra are normalized to the integrated C 1 s signal.

surface-sensitive method based on the oxidation of chemisorbed CO to determine the ESA of the catalysts [63,64]. Cyclic voltammograms of the studied materials recorded under CO atmosphere are presented in Fig. 7c. Both the CO desorption peak area and the corresponding ECSA decrease across the series p1 > p2 > p3 ≈ p4, with ECSA values derived from hydrogen and CO analysis being closely comparable (Fig. 7b, Table 2).

As the series progresses from p1 → p2 → p3 → p4, both the intensity and shape of the CO desorption peak change, with the peak shifting towards lower potentials and exhibiting an increasingly resolved multiplet structure (Fig. 7c). This evolution indirectly indicates a progressive chemical heterogeneity in surface composition, and/or nanocrystal structure, and/or size distribution across the p1–p4 series [65]. The systematic decrease in the CO oxidation onset potential over this series indicates enhanced catalytic activity for CO oxidation.

$ECSA_H$ and $ECSA_{CO}$ of JM40 are comparable to those of catalysts p3 and p4. The CO desorption region in the CV of JM40 exhibits two maxima, with the first weakly pronounced peak occurring at approximately 0.72 V (Fig. 7c). The most intense and clearly resolved CO desorption maximum for JM40 occurs at approximately 0.83 V (Fig. 7c). A two-peak shape of the CO desorption voltammogram is also observed in materials p2, p3, and p4, and when moving from p2 to p4, the presence of two current maxima becomes more pronounced (Fig. 7c), which may be associated with the enrichment of the NP surfaces with platinum across the series of Pt_3Pd/C catalysts p1–p4. The distinct CO oxidation behavior of the bimetallic catalysts prompted a comparison of the electrochemical performance of the least (p1) and most (p4) active catalysts in methanol oxidation, presented in the Supplementary Material (Figure S4). This comparison indicated that catalyst p4 exhibits superior activity in this reaction compared to material p1 (Figure S4, Supplementary Material). It should be noted that the results obtained in studying the CO and methanol oxidation reactions indicate a clear difference between platinum-palladium catalysts, but do not unambiguously explain the reasons for such differences, since they do not allow one to separate the contributions of surface composition, particle size and structure, as well as other structural factors.

The catalytic activity of bimetallic Pt_3Pd and commercial Pt/C catalysts towards oxygen reduction is evaluated by analyzing linear sweep voltammograms recorded at various disk electrode rotation rates (Fig. 7d). From the slopes of the Koutecký–Levich plots (Fig. 7e), it was determined that approximately 4 electrons are involved in the ORR for all investigated catalysts. The 4-electron mechanism corresponds to the direct reduction of oxygen to water. The extrapolation of the Koutecký–Levich plots to the ordinate axis at E = 0.90 V (Fig. 7e) allows for the calculation of mass (I_{mass}) and specific (I_{sp}) activities, which increase across the series p1 < p2 < p3 ≤ p4 (Table 2).

It has been well established that the mass activity of a catalyst is determined by: (1) the electrochemically active surface area; (2) the contribution of different nanoparticle surface sites, such as facets, edges, vertices, and steps, to the total active surface; and (3) the intrinsic activity of individual sites. This relationship can be expressed as follows [66]:

$$I_{mass} = \sum_{0}^{k} (\Theta_k * I_k^{sp}) * ECSA \quad (1)$$

where Θ_k is the fraction of surface sites characterized by intrinsic activity I_k^{sp}.

Since ECSA decreases across the series p1 → p2 → p4 ≈ p3, Eq. (1) indicates that the observed increase in mass activity towards ORR is driven by an enhancement of the intrinsic catalytic activity of the metallic surface (compare the values of I_{sp} in Table 2). This enhancement may arise from either (1) an increased population of active crystallographic facets due to morphological changes or (2) an enhanced intrinsic facet activity resulting from an altered surface composition in bimetallic nanoparticles. Unfortunately, the small nanoparticle size precludes a direct morphological assessment using TEM.

Enhancement of platinum catalytic activity in the ORR through alloying with d-metals has been theoretically well-justified [67] and confirmed by extensive experimental studies. The primary mechanisms include modification of the lattice parameter of the alloy relative to pure platinum and alteration of d-orbital energies, which both favorably affect oxygen and hydroxyl group adsorption on nanoparticle surfaces [67,68]. The most effective promoters of platinum are Co, Ni, Cu, and Fe [68]. The distinct advantage of Pt–Pd alloying, which causes minimal changes in the lattice parameter, lies in two factors: (1) palladium exhibits superior thermodynamic stability compared to other d-metals, and (2) if Pt_xPd/C catalysts achieve comparable activity to Pt/C, it would effectively expand metal reserves for electrocatalyst production [28,69]. Critically, Pt_xPd/C catalysts must also match Pt/C durability.

Catalyst durability is assessed by monitoring changes in the ECSA and the ORR mass activity after accelerated stress testing (the conditions are presented in the Experimental section). The cyclic voltammograms recorded after the stress-test are presented in Figure S5. The quantitative

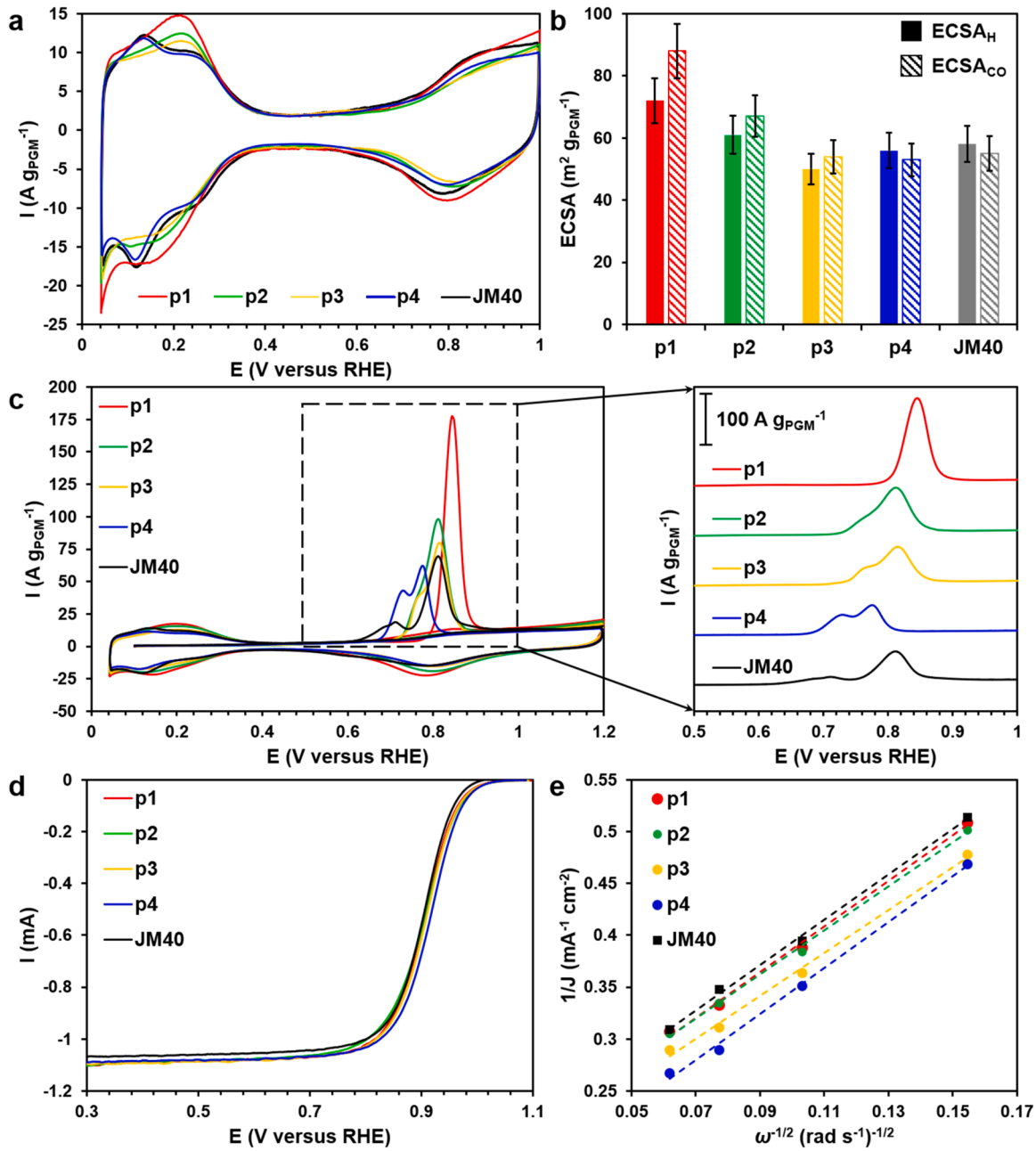

Fig. 7. Cyclic voltammograms recorded under (a) Ar atmosphere and (c) CO atmosphere; (b) electrochemically active surface areas calculated from atomic hydrogen and CO adsorption/desorption; (d) linear sweep voltammograms for oxygen reduction at 1600 rpm disk rotation speed; (e) Koutecky–Levich plots for investigated catalysts. Conditions of measurements are described in the Experimental section.

electrochemical parameters before and after the stress testing are compiled in Table 2.

Accelerated stress testing caused systematic degradation across all catalysts, with the hydrogen and oxygen regions experiencing a marked decrease in CV currents (Figure S5 a, c, e, g, i). This is accompanied by a corresponding decline in ECSA (Table 2).

The shape of the CVs of the Pt_3Pd/C catalysts p1, p2, and p3 changes after stress testing: the hydrogen regions now exhibit two peaks corresponding to strongly and weakly bound hydrogen, as observed both before and after stress testing for the Pt/C and p4 samples (Figure S5a, c, e, g, i). This change in the appearance of the hydrogen region indicates enrichment of the nanoparticle surface with platinum. AST induces changes in the shape and position of the CO oxidation peaks, along with a decrease in ECSA (Table 2), as determined by CO-stripping voltammetry (Figure S5 b, d, f, h, j). Catalyst p1 undergoes the most pronounced changes (Figure S5b): the shape and position of the CO oxidation peak become similar to those of catalysts p4 and JM40, indicating a clear change in the size, degree of agglomeration, and surface chemical composition of the nanoparticles. In general, this trend is also observed for the p2 and p3 samples (Figure S5d, f). For the p4 and JM40 materials, only a decrease in the CO oxidation peak currents is observed (Figure S5h, j), which indicates a smaller loss of ECSA (Table 2) associated with a moderate increase in NP size and, in the case of p4, a minimal change in NP composition.

Given the moderate precision inherent in mass activity determination, small differences in mean values should be treated cautiously; however, the ORR activity ranking post-AST is approximately: p4 > p3 > p1 > p2 ≥ JM40 (Table 2). Notably, ECSA and mass activity degradation exhibit imperfect correlation. This differential degradation behavior is likely due to both the distinct atomic distribution of Pt/Pd

Table 2
Parameters characterizing the electrochemical performance of the catalysts before and after AST.

Sample	$ECSA_{CO}$, m^2 g_{PGM}^{-1}	$ECSA_{CO}$, m^2 g_{Pt}^{-1}	I_{mass}, A g_{PGM}^{-1}	I_{mass}, A g_{Pt}^{-1}	I_{sp}, A m_{PGM}^{-2}
p1	88 ± 9	104 ± 10	280 ± 28	331 ± 33	3.2 ± 0.6
p1 after AST	48 ± 5	57 ± 6	245 ± 25	290 ± 29	5.1 ± 1.0
p2	67 ± 7	78 ± 8	295 ± 30	345 ± 35	4.4 ± 0.8
p2 after AST	46 ± 5	54 ± 5	210 ± 21	246 ± 25	4.6 ± 1.0
p3	56 ± 6	65 ± 7	325 ± 33	377 ± 38	5.8 ± 1.2
p3 after AST	41 ± 4	48 ± 5	254 ± 25	295 ± 30	6.2 ± 1.2
p4	53 ± 5	63 ± 6	411 ± 41	485 ± 49	7.8 ± 1.6
p4 after AST	37 ± 4	44 ± 4	271 ± 27	320 ± 32	7.3 ± 1.4
JM40	55 ± 6	55 ± 6	276 ± 28	276 ± 28	5.0 ± 1.0
JM40 after AST	38 ± 4	38 ± 4	205 ± 21	205 ± 21	5.4 ± 1.0

within individual bimetallic nanoparticles and variations in the structural properties of the catalytic layer across the electrode samples. Crucially, after AST, the Pt_3Pd/C catalysts retain ECSA and mass activity levels that are at least as high as, and in some cases (especially p4) higher than those of the commercial JM40 reference. It is important that after AST, the specific activity of the p4 material remains high, significantly exceeding that of the other catalysts (Table 2), which indicates the high stability of the composition and architecture of the bimetallic nanoparticles.

The ambiguous results of measurements performed after accelerated stress testing of the catalysts, as well as the need to select one of the Pt_3Pd/C samples for long-term life testing at the MEA, necessitated additional TEM and EDX studies of samples p1 and p4 after AST. Fig. 8 shows TEM images of the surface fragments of catalysts p1 and p4 after AST, as well as the corresponding histograms of the NPs size distribution. For material p1, stress testing leads to an increase in the nanoparticle size and the appearance of a significant number of agglomerates, while for material p4 it leads to an insignificant change in the particle size and size distribution (compare the histograms for the catalysts before and after the stress test in Fig. 8c and f). The thermodynamic stability of NPs, as well as the area of their contact with the support surface, and consequently, adhesion to this surface, increase with nanoparticle size. Therefore, degradation of a catalyst with large NPs should decrease regardless of the prevailing degradation mechanism: both due to hindered migration and aggregation of nanoparticles, and due to a decrease in the dissolution rate of platinum. Since the fraction of small-sized nanoparticles in catalyst p4 is initially small, the increase in the fraction of large-sized nanoparticles as a result of platinum redeposition (the electrochemical Ostwald ripening mechanism) is quite moderate, which is reflected in a weak shift of the size histogram towards larger particles (Fig. 8f)

A study of the compositions of the catalysts and individual nanoparticles compositions after AST showed that the Pd content in sample p1 decreased by approximately threefold, while it remained virtually unchanged in sample p4 (Fig. 9). This is confirmed by both EDX mapping and line scanning of the nanoparticle composition (Fig. 9c, d, and g, h, respectively). The insignificant effect of AST on the average size of bimetallic Pt_3Pd nanoparticles and the unchanged composition of the p4 material indirectly confirm the presence of stable "Pt-rich shell - Pd-rich core" architecture in most nanoparticles. Thus, the high stability of p4 catalyst is attributed to a combination of the large size of the nanoparticles and their more stable structure.

The electrochemical measurements, combined with compositional and microstructural characterization, demonstrate that the reduction state of the platinum precursor medium at the moment of $PdCl_2$ introduction profoundly influences the properties of the resulting Pt_3Pd nanoparticles. This influence is complex and multifaceted. The greater the extent of $[PtCl_6]^{2-}$ reduction at the moment of $PdCl_2$ addition, the larger the resulting nanoparticles. ECSA values of the Pt_3Pd/C catalysts correlate well with mean nanoparticle size; however, ORR activity increases counterintuitively across the series p1 ≤ p2 < p3 < p4, opposite to the ECSA trend. These results indicate that Pt_3Pd nanoparticles in the synthesized catalysts differ in nanocrystal structure, surface composition, or both. Importantly, most synthesized samples match or exceed commercial Pt/C in both ORR activity and durability under AST conditions. Their superiority appears even more pronounced when comparing the characteristics of Pt_3Pd/C and Pt/C on a platinum-mass basis (Table 2).

To further evaluate catalyst performance, a comparison of the bimetallic Pt–Pd catalysts with commercial Pt/C in membrane-electrode assemblies of hydrogen-air fuel cells was carried out. Moreover, based on the results of a comprehensive study of electrochemical behavior, as well as the size, composition, and structure of the nanoparticles after AST, the Pt_3Pd/C catalyst p4 was selected for resource testing in the MEA of the PEMFC.

3.5. DFT and Microkinetic Modeling of the Oxygen Reduction Reaction on Pt-Pd Nanoparticles

To gain deeper insight into the origin of the enhanced catalytic activity observed experimentally, particularly the increase in mass-specific activity for ORR across the series p1 → p4 (Section 3.5), we performed a DFT study of four ~1 nm, 79-atom nanoparticle models representing architectural motifs that could arise from our kinetically controlled synthesis: Pt_{79} (monometallic platinum, baseline), $Pd_{19}@Pt_{60}$ (Pd core with a pure Pt shell, a perfectly segregated architecture resembling p1), $Pd_{19}@Pt_{53}Pd_7$ (Pt shell lightly doped with Pd), and $Pd_{19}@Pt_{30}Pd_{30}$ (mixed Pt-Pd shell near 1:1 ratio), the latter two more closely resembling the mixed-surface composition suggested by XPS and MD simulations for p3/p4.

For each model, we first calculated the adsorption of an O_2 molecule in the end-on (Pauling) and side-on (Griffiths) configurations at over 200 sites per particle (Supplementary Material, Section 3.1, Figure S10, Table S5). These calculations show that alloying with Pd modulates the electronic interaction with O_2: in the catalytically relevant Griffiths configuration, the bimetallic particles transfer more charge to the antibonding O_2 orbitals and bind O_2 more strongly than pure Pt_{79}. However, this trend alone does not explain the experimental activity ranking: $Pd_{19}@Pt_{60}$, structurally closest to p1 (the least active catalyst), shows the strongest O_2 binding and the largest charge transfer among all four models, higher even than the mixed-shell $Pd_{19}@Pt_{53}Pd_7$ and $Pd_{19}@Pt_{30}Pd_{30}$ models that more closely resemble the more active p3/p4 catalysts. Adsorption energetics for a single O_2 molecule are therefore an incomplete descriptor of ORR activity: they omit the proton-electron transfer steps and the OOH*, O*, and OH* intermediates that govern the reaction under working conditions.

To address this directly, we extended the DFT dataset with the adsorption energies of O*, OH*, and OOH* at every distinct site (top, bridge, and hollow) of the four nanoparticle models (2272 relaxed structures in total; Supplementary Material, Section 3.2, Figures S12, S13), and combined the results into a computational-hydrogen-electrode-based microkinetic model of the four-electron ORR, comprising the associative pathway together with a competing, chemically activated O-O bond-cleavage channel (Supplementary Material, Sections 3.3–3.11). Reaction barriers, thermochemical corrections, and rate prefactors were transferred from literature without fitting to the present measurements; only 53 local ensembles for which all four intermediates were available on the same set of metal atoms were retained

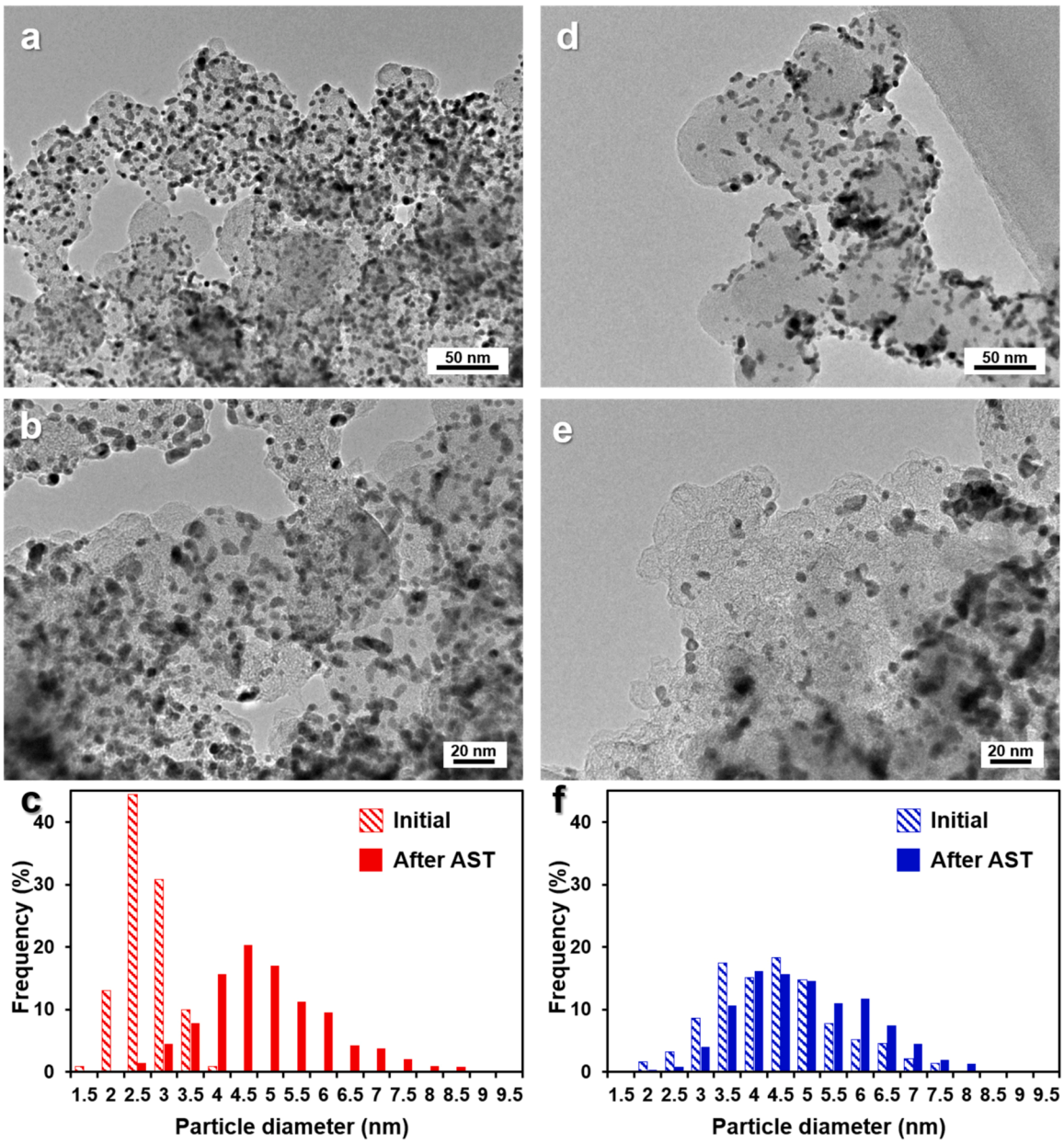

Fig. 8. TEM images of the p1 (a, b) and p4 (d, e) catalysts after AST, and NPs size distribution histograms for these catalysts before and after AST (c, f). The average NP size after AST is 4.9 nm for p1 and 5.0 nm for p4.

and treated as independent, equally weighted site pools. Steady-state rates, surface coverages, and the degree of rate control of each elementary step were then obtained at potentials between 0.10 and 0.92 V vs RHE (Fig. 10).

At 0.90 V, degree-of-rate-control analysis identifies removal of the last adsorbed oxygen intermediate, OH*, rather than O_2 activation, as the step controlling the overall rate for Pt_{79}, $Pd_{19}@Pt_{60}$, and $Pd_{19}@Pt_{53}Pd_7$ (degree of rate control 0.52, 0.49, and 0.79, respectively); only for $Pd_{19}@Pt_{30}Pd_{30}$ does the first proton-coupled electron transfer (O_2* → OOH*) become dominant (0.78) (Fig. 10e). Direct O-O bond cleavage never controls the rate for any of the four models (degree of rate control ≤0.01), even though it carries a substantial share of the flux for Pt_{79} (40%) and $Pd_{19}@Pt_{53}Pd_7$ (68%) (Fig. 10d). Consistent with this, the calculated surface coverages at 0.90 V are dominated by OH* and O* on all four particles, and $Pd_{19}@Pt_{53}Pd_7$ is essentially fully poisoned by O* (coverage ≈ 1); only $Pd_{19}@Pt_{30}Pd_{30}$ retains a noticeable fraction of vacant sites (Supplementary Material, Figure S14).

Resolving the local ensembles individually reconciles this picture with the O_2-adsorption trend above. When Pd is present within the first coordination shell of the adsorption site, it stabilizes O* by about 0.5 eV relative to Pt-only ensembles, and at this level of stabilization the ensemble is effectively poisoned: within $Pd_{19}@Pt_{30}Pd_{30}$, ensembles without a directly coordinating Pd atom carry essentially the entire calculated rate (>97%), while chemically identical ensembles in which Pd directly binds the adsorbate are many orders of magnitude slower (Supplementary Material, Figure S16). Conversely, Pd atoms that remain subsurface, without directly coordinating the adsorbate, promote the surrounding Pt-only ensembles: $Pd_{19}@Pt_{60}$, whose shell is pure Pt with all Pd buried in the core, is calculated to be markedly more active at comparable O* binding than pure Pt_{79}. Site-resolved analysis of the original O_2-adsorption dataset itself (Supplementary Material, Figure S11) shows a related but mechanistically distinct trend: at matched site geometries, the Pt-only sites of all three bimetallic particles bind O_2 more strongly, on average, than the equivalent sites on pure Pt_{79}

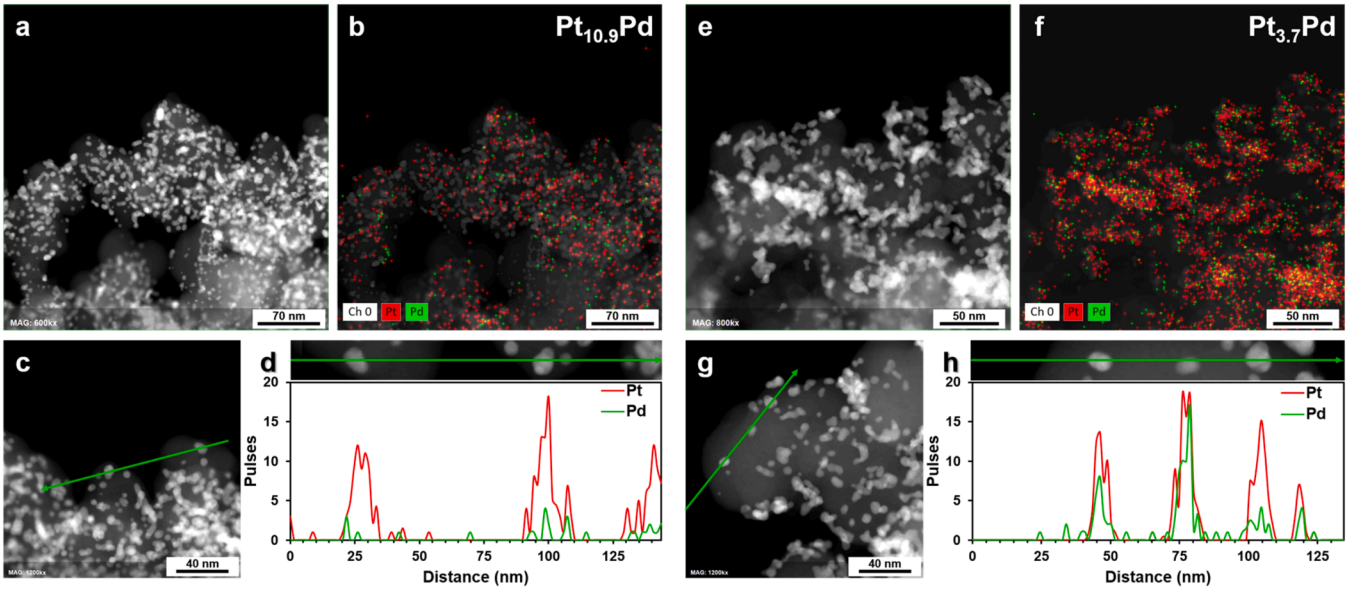

Fig. 9. HAADF-STEM images (a, c, e, g) and EDX elemental mapping of Pt and Pd (b, f); line-scanning of individual NP and HAADF-STEM images of the corresponding NPs (d, h) after AST. Materials: p1 (a-d) and p4 (e-h).

(Pt-only-site averages of −0.49, −0.50, and −0.58 eV for $Pd_{19}@Pt_{60}$, $Pd_{19}@Pt_{53}Pd_7$, and $Pd_{19}@Pt_{30}Pd_{30}$, respectively, versus −0.38 eV for Pt_{79}), showing that this is a general effect of Pd content on the surrounding Pt lattice rather than a feature unique to the fully segregated $Pd_{19}@Pt_{60}$ architecture. Because none of $Pd_{19}@Pt_{60}$'s sites touch Pd directly, however, none are diluted by a separate, Pd-adjacent population, which is why its particle-averaged O_2 binding comes out strongest overall (Table S5), not because of a mechanism special to it. This single-molecule descriptor cannot, on its own, predict which of these energetically favorable sites go on to support fast ORR turnover: that requires resolving the full O*/OH*/OOH* dataset and the site-level microkinetic model, where direct coordination of the reaction intermediates, not merely O_2 itself, is what determines whether a site is promoted or poisoned.

This model is a semiquantitative, mechanistic tool rather than a predictor of the polarization curve: kinetic prefactors and proton-coupled electron-transfer barriers are transferred from the literature, local ensembles are weighted equally in the absence of measured site populations, and solvation, the double-layer field, lateral interactions, and potential-dependent restructuring are not included (Supplementary Material, Section 3.10). We therefore do not use the model to rank the four particles quantitatively or to claim a one-to-one correspondence with catalysts p1-p4; its value lies in the qualitative, mechanistic picture it provides. Within this picture, ORR activity is governed chiefly by the strength of O*/OH* binding rather than by O-O bond activation, and palladium can either promote or poison a given site depending on whether it lies beneath the adsorption ensemble or coordinates it directly.

This mechanistic picture is consistent with the gradient, Pt-enriched but Pd-containing surface architecture indicated by XPS and by the MD simulations of Section 3.1: in such a surface, most adsorption ensembles are expected to be Pt-only or have Pd only in a subsurface position, which the model predicts to be more active than ensembles with directly Pd-coordinated sites. The kinetically controlled synthesis strategy explored in this work, by tuning the extent to which palladium is exposed at versus buried beneath the nanoparticle surface, therefore provides, in principle, an additional degree of control over the balance between these two competing effects of palladium on the oxygen reduction reaction.

3.6. Performance of Pt_3Pd/C catalysts in membrane-electrode assemblies of hydrogen-air fuel cells

The Pt_3Pd/C and commercial JM40 catalysts are evaluated as cathode materials in membrane-electrode assemblies (MEAs) initially at 100% RH, then at 40% RH of inlet gases. The polarization curves and corresponding power density profiles at 100% RH are remarkably similar for the commercial Pt/C and synthesized Pt_3Pd/C catalysts, with the notable exception of p3 (Fig. 11a), which demonstrated somewhat reduced activity.

This is a highly positive result, demonstrating that even without optimization of the ink composition and deposition conditions, the synthesized Pt_3Pd/C catalysts perform comparably to commercial Pt/C.

Even more impressive results emerged from measurements conducted at 40% inlet gas RH, which more closely approximates real-world fuel cell operating conditions with open cathodes: MEAs based on catalysts p1 and p4 exhibited notably superior performance relative to JM40 (Fig. 11b). Significantly, the p1-based MEA showed lower overpotential across the entire voltage range (Fig. 11b). Maximum power density at low RH increases systematically across the series p3 ≤ p2 < JM40 < p4 < p1 (Fig. 11b, c), with the p1 MEA power density exceeding that of the commercial Pt/C MEA by 19%.

Based on the collected data, the Pt_3Pd/C catalyst p4 was selected for additional life tests in fuel cell prototypes. A commercial Pt/C analogue, JM40, was tested as a comparison sample. The membrane-electrode assemblies were tested for 30,000 cycles in accordance with the recommendations of the US Department of Energy [70,71]. Fig. 12 shows the changes in the I–V characteristics during life tests of the p4 and JM40 materials.

Despite initially similar power characteristics, the p4-based MEA degrades significantly more slowly than the JM40-based MEA (Fig. 12). After 30,000 testing cycles, the maximum specific power of the JM40-based MEA was 400 mW cm^{-2} (~ 1000 W g_{PGM}^{-1}), while when using p4 on the cathode, it was approximately 510 mW cm^{-2} (~ 1275 W g_{PGM}^{-1}), a 28% increase. The specific power values of the MEAs decreased by 41% and 26%, respectively (Fig. 12).

The differences in the stability of the catalysts studied are confirmed by electrochemical impedance spectroscopy data. Fig. 13 a, b show the spectra for membrane electrode assemblies (MEAs) with p4 and JM40 cathodic catalysts, respectively, measured at 0.8 V (i.e., in the activation-loss region).

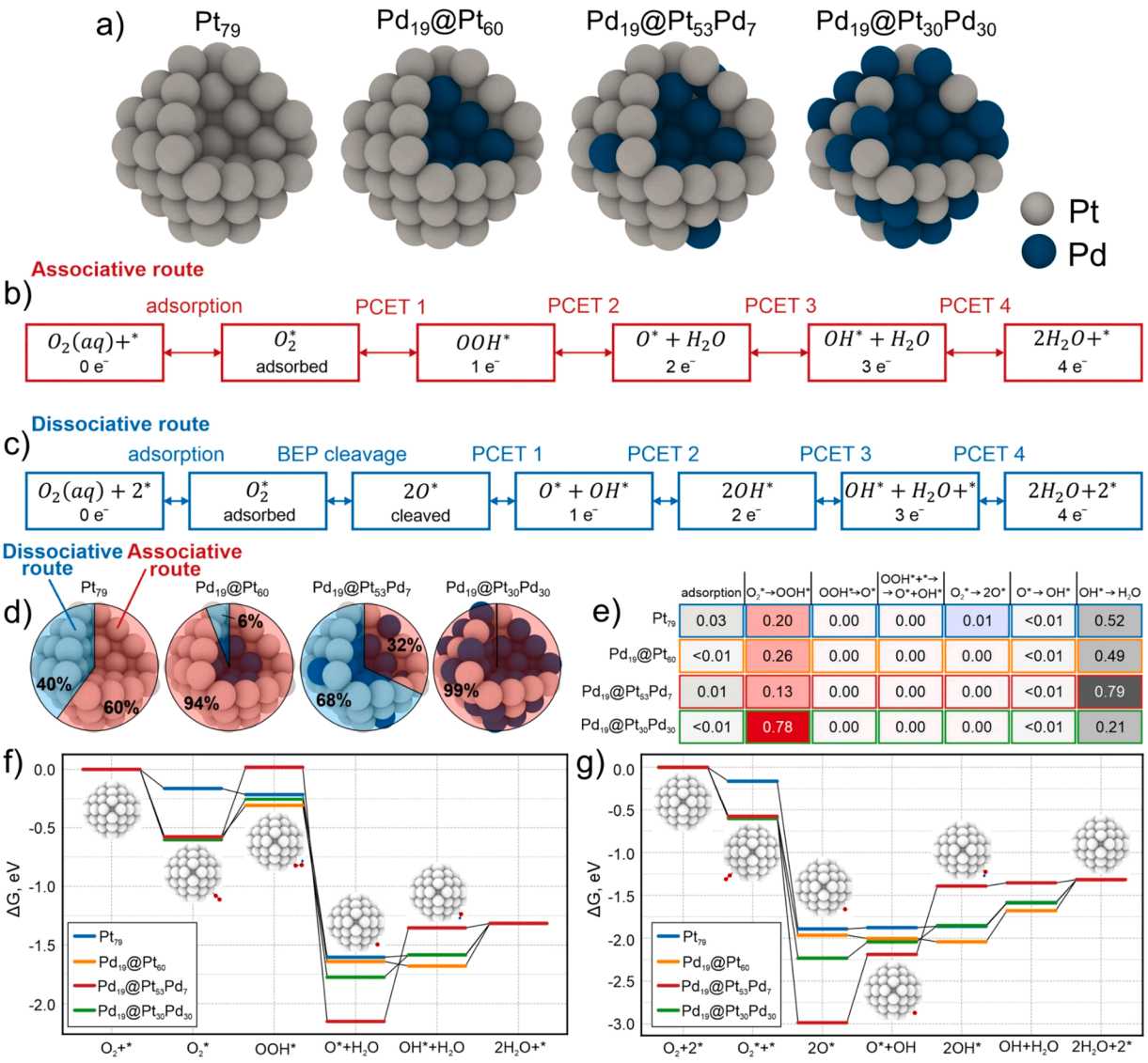

Fig. 10. Atomic structures of the considered bimetallic nanoparticles and the two ORR routes on the Pt-Pd ensembles from the microkinetic model at 0.90 V vs RHE. (a) Atomic structures of the four considered bimetallic nanoparticle models (Pt_{79}, $Pd_{19}@Pt_{60}$, $Pd_{19}@Pt_{53}Pd_7$, $Pd_{19}@Pt_{30}Pd_{30}$). (b) Associative and (c) dissociative routes, differing in whether the O-O bond is broken by successive proton-coupled electron transfers or cleaved chemically after adsorption. (d) Partitioning of the O_2 flux between the two routes for each particle, averaged over all local ensembles of that particle. (e) Degree of rate control (dimensionless) of the elementary steps; cells marked "< 0.01" denote steps that do not control the rate. (f, g) Free-energy profiles along the two routes, averaged over the same ensembles and referenced to O_2 in solution; insets show the corresponding intermediates.

The equivalent circuit used to fit the experimental data is shown in Fig. 13 a. The series resistance R_{ohm} represents the ohmic resistance of the cell and is connected in series with the two R||CPE branches. The $R_{ct}||CPE_{dl}$ element describes the main arc in the spectra, which is associated with the charge-transfer process at the cathode, whereas the $R_{pt}||CPE_{pt}$ element characterizes the high-frequency region of the spectra, which is governed by proton transport in the cathodic catalyst layer. Before stress testing, the p4-MEA exhibited a 19% higher cathode charge-transfer resistance (R_{ct}) than the JM40-MEA. However, after stress testing, the R_{ct} for the JM40-MEA was 90% higher than that for the p4-MEA (Fig. 13 c). Since R_{ct} reflects the ORR kinetics at the MEA cathode, the observed trend indicates a substantially higher stability of the p4 cathode catalyst in terms of catalytic activity. At the same time, no significant change is observed in R_{ohm} — a parameter primarily associated with the resistance of the proton-exchange membrane — during stress testing. This indicates that there is no substantial membrane degradation and that such degradation is not the cause of the observed differences in the performance characteristics of the MEAs (Fig. 13 d).

4. Conclusion

A series of carbon-supported platinum–palladium electrocatalysts p1–p4 was synthesized via kinetically controlled polyol synthesis with real-time monitoring of precursor reduction. The $[PtCl_6]^{2-}$ and $PdCl_2$ precursors were reduced both simultaneously (p1) and sequentially (p2–p4), the latter with $PdCl_2$ introduced at programmed moments corresponding to different "depths" of Pt(IV) reduction. The "depth" of Pt(IV) reduction was determined by continuous monitoring of the state of the reaction medium in real time.

The precious-metal loading in the synthesized catalysts is close to 40 wt%, and the composition, $Pt_{3.4\pm0.4}Pd/C$, was found to be close to the nominal 3:1. Moreover, virtually all nanoparticles are bimetallic, i.e., they consist of both platinum and palladium atoms. At the same time the timing of $PdCl_2$ introduction had a significant effect on the size, structure, and electrochemically active surface area of the nanoparticles: an

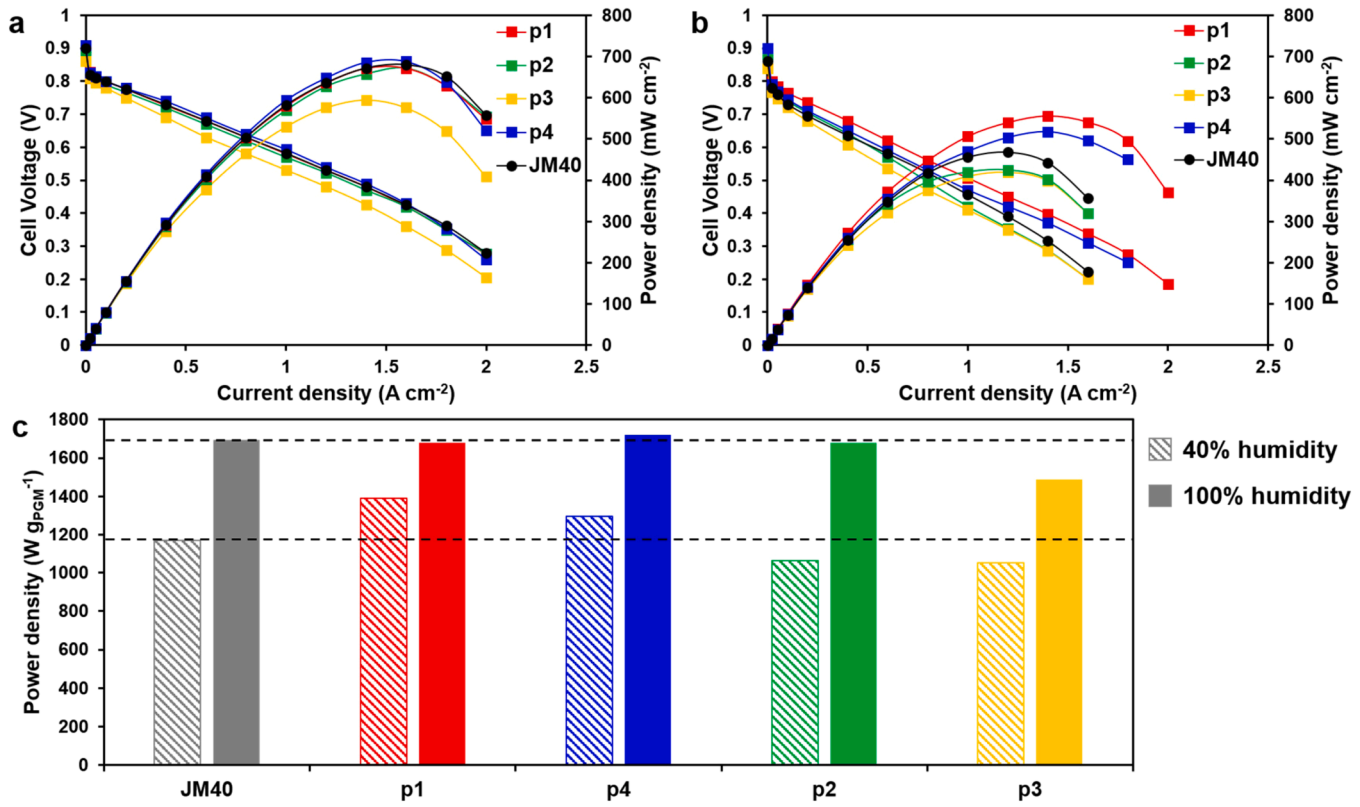

Fig. 11. Fuel cell polarization curves and specific power density vs. current density for MEAs with cathodes based on studied catalysts at 100% (a) and 40% (b) inlet gas RH, with corresponding maximum power density values (c).

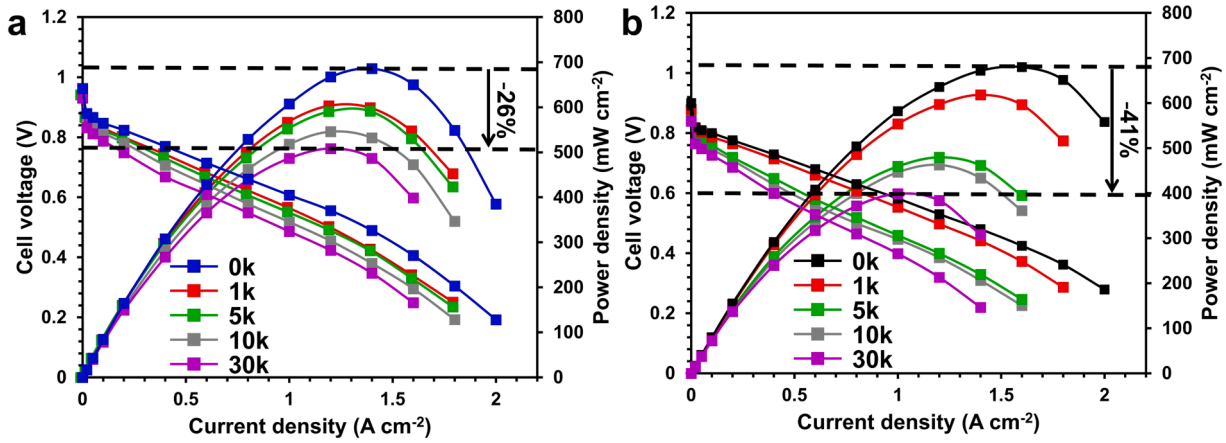

Fig. 12. Fuel cell polarization curves and specific power density vs. current density for MEAs with cathodes based on p4 (a) and JM40 (b) catalysts at 100% inlet gas RH before (0k) and after 1k, 5k, 10k and 30k cycles of stress test. The dotted lines indicate the values of maximum power for the initial state and after the stress test.

increase in the introduction time (p1 → p4) led to an increase in nanoparticle size and a decrease in the ECSA of the $Pt_{3.4\pm0.4}Pd$/C catalysts.

Because of the small nanoparticle size (2.7–4.4 nm), the radial distribution of Pt and Pd could not be resolved directly: the EDX line profiles of individual nanoparticles reveal a sharp Pt/Pd concentration gradient at the particle periphery but cannot show how the components are distributed inside the particles. Nevertheless, a set of independent observations, obtained by methods that probe different aspects of the nanoparticle structure (kinetics of formation, atomistic modeling, near-surface composition, electrochemical response of the outermost atomic layers, and behavior under potential cycling), is consistently explained by a "Pd-rich core – Pt-rich shell" architecture of the $Pt_{3.4\pm0.4}Pd$ nanoparticles:

- according to RGB and redox-potential monitoring, Pd(II) is reduced and nucleates almost immediately after $PdCl_2$ addition, whereas Pt (IV) is reduced slowly via Pt(II). Palladium nuclei therefore form first, and platinum is deposited onto them The line profiles show the first characteristic feature of core–shell particles, a pronounced Pt/Pd concentration gradient at the nanoparticle periphery.
- XPS shows that the near-surface layers of both p1 and p4 are substantially enriched in platinum relative to the bulk composition, with Pt and Pd in the metallic state;
- the hydrogen region of the CV of the p4 catalyst has the same shape as that of Pt/C and differs from that of the p1–p3 catalysts. Only after the accelerated stress test do the CVs shapes of the $Pt_{3.4\pm0.4}Pd$/C catalysts p1–p3 match that of Pt/C;

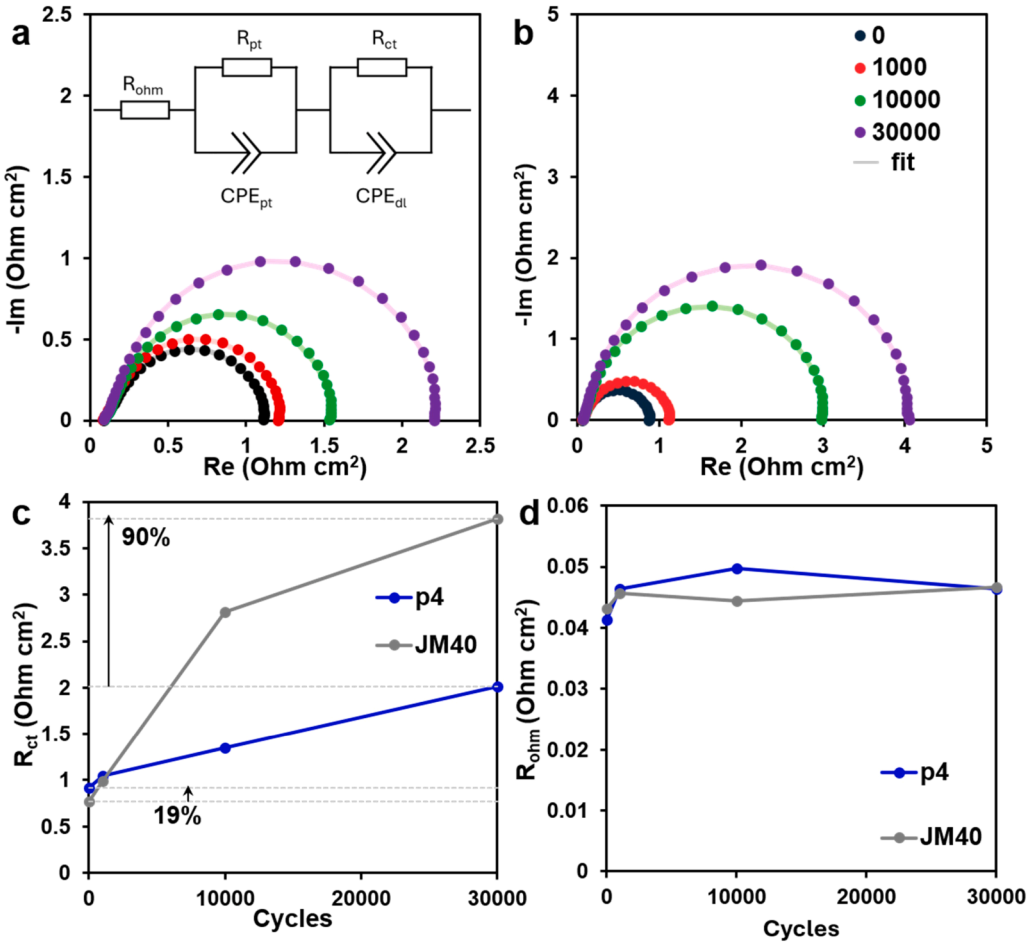

Fig. 13. Electrochemical impedance spectra measured at 0.8 V for MEAs with p4 (a) and JM40 (b) cathode catalysts, where (a) includes the equivalent circuit used for fitting the spectra; c) evolution of the cathode charge-transfer resistance (R_{ct}) during MEA stress testing; d) evolution of the ohmic resistance (R_{ohm}) during MEA stress testing.

- despite the decrease in ECSA, the mass activity of the platinum–palladium catalysts in the ORR increases with increasing time of palladium precursor introduction into the reaction medium, i.e., from p1 to p4. The p4 has the highest specific and mass ORR activity (I_{sp} = 7.8 A m^{-2}, I_{mass} = 411 A g_{PGM}^{-1}) and the highest activity in methanol oxidation compared with p1. This activity is retained after AST (I_{sp} = 7.3 A m^{-2}). In the same p1–p4 series, the chemisorbed CO oxidation peaks shift regularly toward lower potentials;
- after AST the Pd content in p1 decreased by about threefold, which shows that a substantial part of Pd in p1 is accessible to dissolution, whereas in p4 the Pd content and the particle size distribution remained practically unchanged. The Pd in p4 is therefore effectively screened by the Pt-rich shell. Accordingly, p4 retains the highest ORR activity after AST.

Molecular dynamics simulations revealed gradient Pt–Pd architectures with Pt-enriched surfaces, which agrees well with the experimental data. DFT calculations combined with microkinetic modeling showed that the ORR activity on these architectures is governed mainly by the removal of adsorbed oxygen intermediates rather than by O_2 activation, and that palladium enhances activity when it is buried beneath the active site but poisons it when it directly coordinates the adsorbate.

Thus, the totality of the experimental and computational data indicates that the differences in the electrochemical behavior of the $Pt_{3.4\pm0.4}$Pd/C catalysts are due not so much to size effects influencing the ECSA as to the surface composition and/or structure of the nanoparticles. Each of the above observations taken separately admits alternative explanations, but a homogeneous alloy, a Pt-core – Pd-shell structure, a mixture of monometallic particles, and a Pd-exposed surface are each inconsistent with at least several of them. The distinctive behavior of the p4 catalyst is most likely due to the highest concentration of platinum atoms in the surface layers of the Pd-rich core – Pt-rich shell nanoparticles.

Overall, the synthesized $Pt_{3.4\pm0.4}$Pd/C catalysts demonstrated a mass activity exceeding that of commercial Pt/C and high durability in accelerated stress tests in an electrochemical cell. The ORR mass activity of the synthesized catalysts also exceeds that of platinum–palladium catalysts reported in the literature (Table S3). In MEA polarization measurements, most samples performed comparably to Pt/C at 100% RH, whereas p4 and p1 showed superior specific power density at 40% RH. In durability tests conducted according to the US DOE protocol, the MEA based on the catalyst p4 delivered a 28% higher maximum power density than the commercial Pt/C-based MEA at the same PGM loading. The observed difference in MEA durability is attributed specifically to the higher degradation resistance of sample p4. Thus, the synthesized $Pt_{3.4\pm0.4}$Pd/C catalysts match or surpass commercial Pt/C (JM40) with identical PGM loading in terms of functional performance, and the p4 catalyst is a very promising candidate for use in MEAs of hydrogen–air PEMFCs.

The effectiveness of the developed kinetic approach, tested on the synthesis of Pt_3Pd/C catalysts, opens up new opportunities for controlling the structure and composition of various nanostructured materials, including multicomponent and high-entropy materials.

Author contributions

V.G. proposed the synthesis method, developed the research plan with K.P., and wrote the manuscript with the participation of all authors. I.K. synthesized the catalysts and performed the bulk of the experimental work on characterizing the materials and studying their electrochemical behavior under K.P.'s guidance. I.Ch., V.B., G.B., A.K. and E. B. performed DFT modeling and interpreted the results. A.K. and K.V. performed the XPS and XRD study of the materials, respectively. I.P. performed the TEM study of the materials and, together with KP, interpreted the results. V.M. and E.B. tested the catalysts in membrane-electrode assemblies of the fuel cell.

Declaration of Competing Interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Vladimir Guterman reports financial support was provided by Russian Science Foundation. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.apcatb.2026.127718.

Data availability

Data will be made available on request.

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